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Charge Injection in Solution-Processed Organic Field-Effect Transistors: Physics, Models and Characterization Methods

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A high-mobility organic semiconductor employed as the active material in a field-effect transistor does not guarantee per se that expectations of high performance are fulfilled. This is even truer if a downscaled, short channel is adopted. Only if contacts are able to provide the device with as much charge as it needs, with a negligible voltage drop across them, then high expectations can turn into high performances. It is a fact that this is not always the case in the field of organic electronics.

In this review, we aim to offer a comprehensive overview on the subject of current injection in organic thin film transistors: physical principles concerning energy level (mis)alignment at interfaces, models describing charge injection, technologies for interface tuning, and techniques for characterizing devices. Finally, a survey of the most recent accomplishments in the field is given. Principles are described in general, but the technologies and survey emphasis is on solution processed transistors, because it is our opinion that scalable, roll-to-roll printing processing is one, if not the brightest, possible scenario for the future of organic electronics.

With the exception of electrolyte-gated organic transistors, where impressively low width normalized resistances were reported (in the range of 10 Ω·cm), to date the lowest values reported for devices where the semiconductor is solution-processed and where the most common architectures are adopted, are ~10 kΩ·cm for transistors with a field effect mobility in the 0.1–1 cm^2/Vs range. Although these values represent the best case, they still pose a severe limitation for downscaling the channel lengths below a few micrometers, necessary for increasing the device switching speed. Moreover, techniques to lower contact resistances have been often developed on a case-by-case basis, depending on the materials, architecture and processing techniques. The lack of a standard strategy has hampered the progress of the field for a long time. Only recently, as the understanding of the rather complex physical processes at the metal/semiconductor interfaces has improved, more general approaches, with a validity that extends to several materials, are being proposed and successfully tested in the literature. Only a combined scientific and technological effort, on the one side to fully understand contact phenomena and on the other to completely master the tailoring of interfaces, will enable the development of advanced organic electronics applications and their widespread adoption in low-cost, large-area printed circuits.

1. Introduction

Organic field-effect transistors (OFETs), based on solution-processed semiconducting conjugated small molecules or polymers, are being adopted in the development of innovative optoelectronics products, such as backplanes for flexible displays and low cost circuits for sensors applications.[1–8] This was possible thanks to a recent fast development in the field[9] and to a corresponding steady improvement in charge carrier field effect mobility, which now exceeds 1 cm^2/Vs for both holes and electrons.[10] Solution-processability has favored their adoption in low-cost, large-area applications, which are expected to revolutionize consumer electronics in the next years, provided that the technology reaches the needed level of maturity in terms of processes standardization, lifetime, and yield.

OFETs are also the subject of intensive studies to improve their performance and meet the requirements for advanced applications, such as RF identification tags and addressing electronics in video-rate displays, where complex digital circuits with demanding performances in terms of clock rate and therefore of switching speed are required.

To an extent there still are both physical and technological issues to be faced. Besides fundamental studies to clarify the nature of the charge carriers and their transport mechanism,[11–13] and the development of materials with improved carriers mobility and environmental stability, it is of crucial importance to find technologically viable approaches for the downscaling of the devices,[14–16] in order to improve the operational frequencies, reduce the required supply voltage, and decrease the device footprint. Reducing the geometrical dimensions of an OFET means shrinking, among other parameters, the channel length *L*, and therefore reducing the channel resistivity; but the carrier injecting/collecting capabilities of the metal/semiconductor interfaces do not scale with *L*, as these are inherently interfacial properties. Obviously, the demand for current (due to the channel) has always to match the supply (due to the contacts), and for optimal device operation, this has to occur with a negligible voltage drop at

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the contacts: in this case contacts are termed ohmic. As downscaling proceeds, the channel gets more and more current hungry, while contacts do not change their properties significantly: if, starting from a certain critical *L*, the demand/supply balance is met with a sizeable voltage drop across the contacts at the expense of the voltage drop on the actual channel, the device becomes contact limited. This has to be avoided, because it results in a current limitation (with respect to the same device with ohmic contacts), which at least partially spoils the downscaling expected benefits. One of the clearest fingerprint of this situation is that the field-effect mobility, extracted from OFET measurements without properly taking into account contact issues, is dependent on *L* and gets lower with shorter channels. This is referred to as apparent mobility, whereas the intrinsic one can be extracted only if the contact-altered dependence of current on applied gate and drain voltages and on *L* is correctly modeled. Besides current reduction, the phenomenology of contact limited OFET includes: non-linear S-shaped output characteristic curves at low drain to source voltage (detrimental for digital applications), and apparent mobilities extracted in the linear regime much lower than saturated mobilities.

It is common practice to quantify and model contact limitations in terms of a contact resistance R_C in series to the accumulated channel. To date the lowest width-normalized contact-resistance values reported for devices where the semiconductor was solution-processed and where the most common architectures are adopted, are ~1–10 $k\Omega \cdot cm$ for transistors with a field effect mobility in the 0.1–1 cm^2/Vs range. This means that, even in the best case, all devices with a channel length shorter than 5 μm suffer contact limitations. What is worse is that it is the evident from the literature that contact optimization is performed on a case-by-case basis, depending on the specific semiconductor, electrodes, and device architecture. While in silicon technologies high local doping in correspondence of the electrodes allows the fabrication of ohmic contacts, in organic electronics such a powerful technique is not available, representing of course a strong technological limitation. Nevertheless, the increasing understanding of the rather complex physical processes at metal/semiconductor interfaces is fostering the development of more general approaches, with a validity that extends to several materials.

This review aims to provide a comprehensive overview of the developments in the field so far, providing the reader with a broad view from fundamentals to technological and experimental techniques.

Section 2 deals with charge injection: we start with the general physics of the metal/semiconductor interface (energy levels alignment in Section 2.1 and current injection in Section 2.2); in Section 2.3 we focus on injection models in OFETs; in Section 2.4 the most widely adopted techniques to tune the injection and reduce resistance limitations are reported, and Section 2.5 describes the effects on R_C produced by particular processing conditions during device fabrication. In Section 3 we aim to provide the reader with a critical overview of the most widely adopted methods to extract R_C in OFETs. In Section 3.1 methods to extract R_C from current–voltage measurements are reported, and in Section 3.2 experimental methods that provide an almost direct measurement of the voltage drops at the contacts are described. Finally, Section 4 reports a survey on the width-normalized contact-resistance values reported in recent literature for both *p*- and *n*-channel solution-processed OFETs.

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2. Charge Injection in OFETs

2.1. Contact Energetics

Generally speaking, when two materials with different Fermi levels (E_F) are brought into contact, they reach thermal equilibrium and establish a common E_F by exchanging carriers. Electrons diffuse from the material with the highest E_F (measured with respect to vacuum) towards the material with the lowest E_F. This gives rise to a contact potential across the interface, equal to the difference of the Fermi levels of the isolated materials. Charges transferred to the semiconductor can be stored in the form of a space charge, as a surface sheet charge (if surface states are present), or a combination of the two. In the first case the contact potential drops in the semiconductor space charge region (and vacuum levels align at the interface), in the second one it drops in the form of a surface dipole at the interface (displacing vacuum levels at the interface).

In the following we summarize the peculiarities of an interface when the two materials brought into contact are a metal

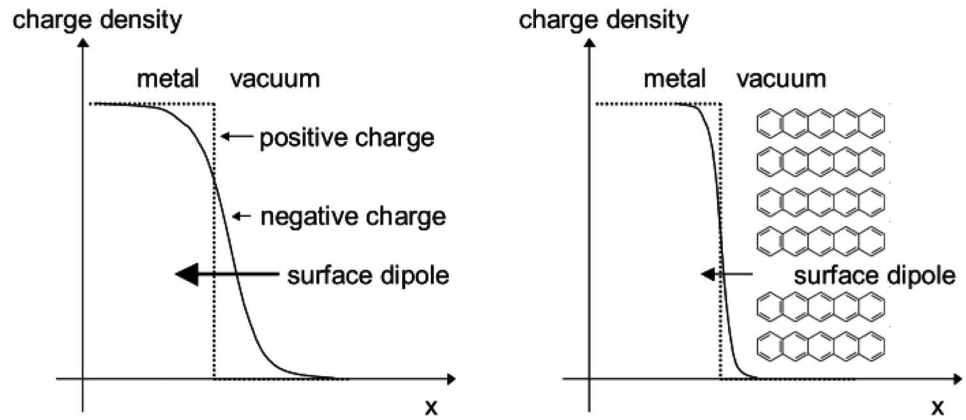

Figure 1. Schematic representation of electron charge density close to a metal surface: on the left, metal electron wavefunctions protrude outside the metal boundary into the vacuum giving rise to a surface dipole; on the right, the presence of a physisorbed molecule pushes back electron tails, and as a consequence the surface dipole and the metal work function diminishes. Redrawn after Ref. [19].

and an organic semiconductor. We provide a simple description of the most relevant physical phenomena, without discussing the underlying experimental techniques involved, most notably photoelectron spectroscopy (PES) and Kelvin Probe (KP): the interested reader is referred to Refs. [17,18] and references therein for a detailed and exhaustive discussion on the subject.

Depending on materials used and on preparation details, different degrees of interaction can occur at a metal/organic interface, ranging from physisorption, where organic molecular orbitals do not hybridize with metal electrons wavefunctions, to chemisorption, with the formation of covalent bonds.[17]

2.1.1. Physisorption

In the relatively simple case of physisorption, typical for solution processing of active materials onto metal layers, which results in passivated metal surfaces, a number of phenomena must be accounted for. First of all it must be recalled that the *metal* work function (W_f) itself is a surface property, and as such is prone to modifcation by the presence of closed-shell organic molecules at the surface. In fact, if one considers an isolated metal, the electron charge density does not abruptly drop to zero at the surface but "leaks" into vacuum, giving rise to a surface dipole contributing to the total work function. When organic molecules are physisorbed at the surface, metal electron tails are pushed back closer to the metal surface by π-electron clouds, thus modifying the surface dipole and ultimately decreasing the work function by as much as many tenths of eV (e.g., a W_f reduction of 0.5–1 eV is reported for gold[18]). This is called the pillow or push-back effect (see **Figure 1**).

In addition, the following physical phenomena occurring at the metal/organic interface are to be taken into account: *i*) screening due to image charges on metal surfaces leads to a reduced transport gap within about the first 2 nm from the interface, as schematically shown in **Figure 2**;[19] *ii*) excess charges in the semiconductor are stored in geometrically and electronically relaxed (polaronic) states; *iii*) due to energetic and positional disorder, HOMO (Highest Occupied Molecular Orbital) and LUMO (Lowest Unoccupied Molecular Orbital) levels are statistically distributed (according to a Gaussian or

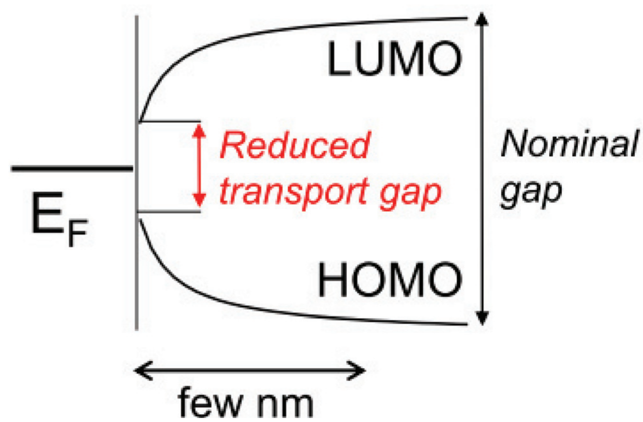

Figure 2. The image charge potential screening at the interface between a metal and a semiconductor results in a local (up to a few nanometers from the interface) reduction of the transport gap (for simplicity, in the figure the metal Fermi level is aligned with the semiconductor midgap, so that no charge transfer occurs at thermal equilibrium). Redrawn after Ref. [19].

an exponential distribution with a width of about 100 meV).[20] Indeed, a model which takes all these three phenomena into account is still missing, even though it is reasonable to expect all of them to play a role.[21] The on-going debate on this issue is beyond the scope of this review; we choose here to describe the interface by taking into account disorder and image charge only, as these will be the key ingredients in our description of carriers injection (Section 2.2), but it has to be said that the same experimental results can be explained equally well by either polaronic states with a well defined energy or by a statistical distribution of states.[21]

We assume that thermal carriers in the semiconductor are negligible (due to relatively large band gaps) and that E_F in the isolated semiconductor lies in the mid of gap, e.g., the semiconductor is intrinsic. Now we focus on the effects of the relative positioning of the metal E_F with respect to the organic semiconductors HOMO and LUMO levels, and for simplicity, we keep the organic semiconductor levels constant and imagine varying

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the metal E_F (**Figure 3**). As a starting point, we consider the isolated metal E_F aligned with the isolated semiconductor E_F. In this case, no charge transfer rigorously occurs at thermal equilibrium and hence no potential drop is expected in the semiconductor. The same holds true as long as the metal E_F departs from semiconductor midgap but is still located well within the transport gap: a common Fermi level is established with a negligible amount of charge transfer, and hence no significant potential drop occurs in the semiconductor. When E_F approaches the LUMO(HOMO) level, sizeable, integer charge transfer from(to) metal to(from) LUMO(HOMO) level takes place, and the stored charge in the semiconductor results in a bending of energetic levels (often improperly referred to as *band* bending), with the LUMO(HOMO) levels acquiring a negative(positive) curvature. The relative positioning of LUMO and HOMO levels with respect to metal E_F right at the interface is assumed to be undisturbed by the charge transfer, because in physisorption regime no surface states, which might give rise to an interface dipole, are expected. The semiconductor voltage drop opposes electron diffusion to(from) the semiconductor and finally establishes thermal equilibrium. Recent simulations[21] and experimental data[18] suggest that sizeable charge transfer occurs when the energy difference between LUMO(E_F) and E_F(HOMO) is less than 0.4–0.6 eV (the smaller value holding for smaller DOS widths). Sizeable level bending (few tenths of eV) occurs on the spatial scale of tens of nm, reducing to less than 10 nm for very small DOS width.[22] Thanks to level bending, the energy difference between LUMO(E_F) and E_F(HOMO) increases on moving away from the interface up to the threshold value of 0.4–0.6 eV: at this relative distance the amount of charge stored in the DOS becomes very small, so that further bending does not occur, at least on the scale of 100 nm, which is the one experimentally explored by PES and KP. The relative positioning of the E_F with respect to HOMO/LUMO right at the interface influences the carrier concentration there, the latter of which gets larger the closer the metal E_F lies to LUMO(HOMO); even larger carrier concentration at the interface occurs when metal E_F lies above(below) LUMO(HOMO). Since far from the interface the relative positioning of E_F with respect to LUMO/HOMO appears to settle at about 0.4-0.6 eV and to be relatively insensitive to the positioning right at the interface, the situation is often referred to as Fermi level pinning regime.

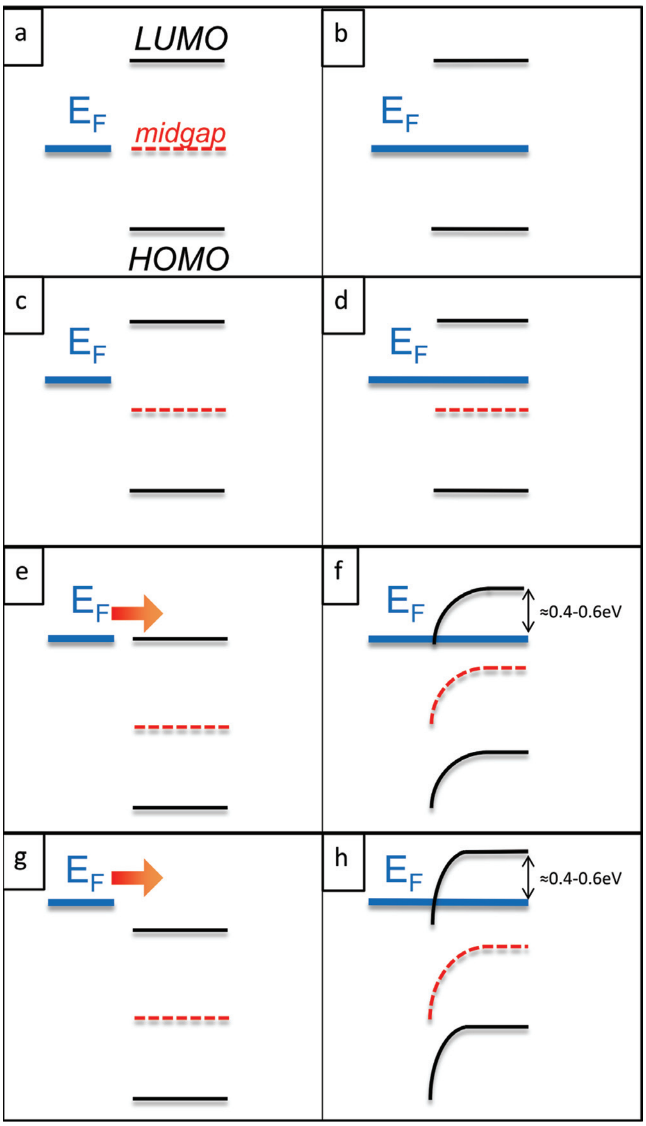

Figure 3. Qualitative sketch of energy levels at metal/semiconductor interfaces in the physisorptive regime: before (panels a, c, e, g) and after (panels b, d, f, h) thermal equilibrium. When the metal Fermi level coincides or is close to the semiconductor midgap (a-d), negligible charge transfer occurs and energy levels are flat. When the metal Fermi level approaches or is above for example the LUMO (e, g respectively), sizeable charge transfer and hence level bending occurs to counterbalance metal to semiconductor electron diffusion (f, h). The LUMO-E_F difference is expected to influence the carrier concentration at the interface (larger in panel h than in panel f). Farther away from the interface the distance between LUMO and E_F increases, carrier concentration drops and energy levels tend to flatten out; the LUMO-E_F difference seems to attain a limiting value of about 0.4 - 0.6 eV, to a first approximation irrespectively of the E_F/LUMO positioning at the interface.

A surface dipole at the interface can arise in case of organic molecules with intrinsic dipoles, if they happen to be at least partially oriented at the interface.

2.1.2. Weak Chemisorption

Weak chemisorption is the next step in the interaction strength scale, typical of very clean ultrahigh vacuum prepared interfaces. In this case, the intimate contact between metal and organic molecules results in metal-hybridization of molecular orbitals, which shifts, broadens and ultimately gives rise to a continuum distribution of states with non-negligible density in what used to be the energy gap. This Induced Density of Interfacial States (IDIS) is filled with the charge of the isolated and neutral molecule up to the so-called charge neutrality level (CNL, **Figure 4**).[23] The relative position of CNL with respect to the metal E_F drives the charge transfer. As shown by DFT-LCAO (Density Functional Theory - Linear Combination of Atomic Orbitals) calculations, the CNL position in the gap is specific to each molecule (and rather insensitive to the chosen metal), and can deviate substantially from the midgap. Considering that permanent dipoles and pillow effect can also play a role, it is clear that the knowledge of the isolated metal work function

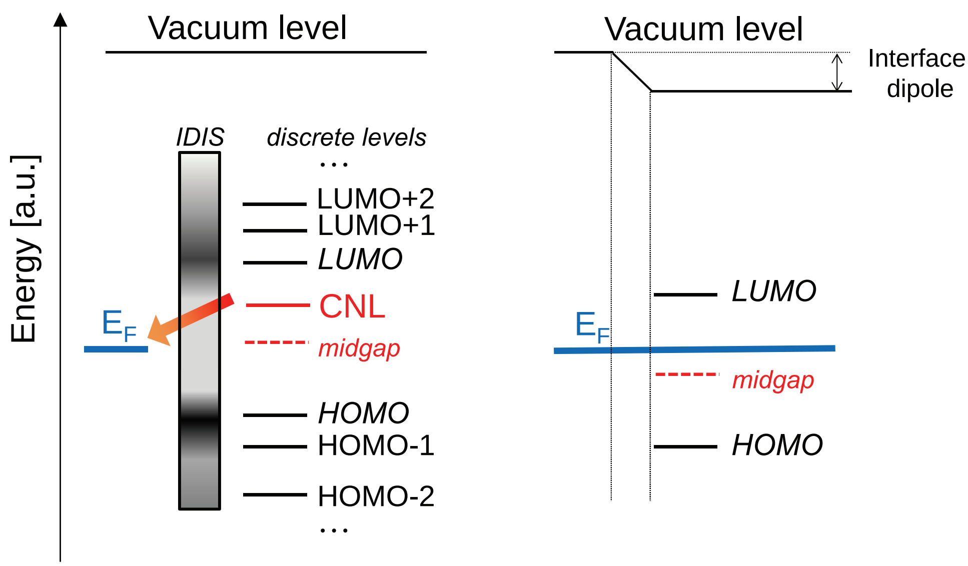

Figure 4. On the left: qualitative sketch of a metal/organic interface where IDIS model applies before thermal equilibrium. The discrete organic energy levels of the isolated molecule (only a few indicated in the figure) broaden due to hybridization with metal wavefunctions. The resulting continuous density of states (darker area means larger density of states) is filled up to the so called Charge Neutrality Level (CNL). Note that due to the broadening being possibly asymmetric for empty and occupied levels, in the general case CNL does not coincide with midgap (in the figure it is shifted towards LUMO). Positioning of CNL with respect to Fermi level determines the extent and direction of charge transfer at the interface: in the figure electron transfer takes place from the organic molecule towards the metal. On the right: the interface at thermal equilibrium has developed an interface dipole.

and of the isolated molecule HOMO/LUMO levels gives but a little information on the final levels alignment at the interface.

2.1.3. Strong Chemisorption

At the highest degree of interaction strength scale we find the case of strong chemisorption, where metal/organic covalent bonding occurs. The situation is typical of clean reactive (transition, alkaline and alkaline earth) metals, and of Self-Assembled Monolayers (SAMs), whose anchoring groups (e.g., thiols) give rise to chemisorption even onto non-reactive metals such as gold. Since chemical bonding can be accompanied by partial charge transfer, an interface dipole may develop. The pillow effect must be taken into account as well due to the close proximity of the chemisorbed molecule and the metal. Finally, in the case of SAM, the molecule end-group can have an intrinsic dipole and hence additional potential energy-step will occur. Non-idealities such as interface roughness and charge trapping at the interface can complicate the picture. Since only the molecules in contact with the substrate are chemisorbed, a possible simplifying view is to consider the metal and the chemisorbed monolayer as a whole non-reactive substrate, and the successive layers as simply physisorbed on top.

To summarize, the energy level (mis)alignment at a metal/organic interface can be hardly predicted from the basic knowledge of the metal Fermi level and of the semiconductor HOMO and LUMO levels. A complete description requires the various interactions, specific to each metal/semiconductor interface, to be properly taken into account. This is not trivial even in the case of ad hoc prepared interfaces, and becomes more complicated in the specific case of OFET source and drain contacts, where device structure, process flow and technological constraints may add a certain degree of uncertainty in the relevant material parameters.

We conclude this section with a brief note on organic/organic (OO) interfaces, as they are becoming technologically relevant as a means to tailor charge injection in OFETs. Firstly, it has to be pointed out that image charge and pillow effects are negligible at such interfaces due to a lower electronic density (if compared to metal/organic interfaces). In most of the cases OO interfaces are dominated by a combination of weak van der Waals dispersion forces, electrostatic interactions and steric repulsions. In case of weak interaction between molecules on opposite sides of the interface (e.g., because of alkyl chains), integer charge transfer can occur if the resulting charge-transfer state is more stable than the neutral ground state. More intimate interaction can result in: *i*) partial charge transfer due to overlap and hence ground state mixing of molecular orbitals (short range interaction, active on the scale of few Å); *ii*) reorganization of the electronic density within the molecules resulting in polarization effects (long range interaction). In both cases a dipole and hence energetic level shift occurs at the OO interface. The phenomena above are very sensitive to the spatial and mutual nanoscale arrangement of molecules at the OO interface: the interested reader is referred to the recent work of Beljonne et al.[24] for more details.

2.2. Charge Injection into Organic Semiconductors

In this section the physics of current injection into organic semiconductors is summarized. The focus will be on peculiarities of disordered organic materials, whereas organic single crystals, which can be modelled within the framework developed for inorganic semiconductors, will not be dealt with.[25]

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As an absolute definition, an ohmic contact acts as an infinite carrier reservoir, thus supplying any current with a negligible voltage drop across itself. Obviously, no such a contact exists in reality. Real contacts can only behave ohmic relative to a certain material, for a certain device topology, and in a certain set of conditions (e.g., bias, temperature, light, etc.).[26] Having clarified this, and taking into account the large variability of carrier mobility of organic semiconductors, and additionally the poor definition of energetic barriers, only qualitative rules can be given for attaining an ohmic contact.

2.2.1. Ohmic Contact: Qualitative Considerations

A one-dimensional problem of electron injection will be considered, with an externally applied voltage V biasing the semiconductor positive with respect to the metal. A general and qualitative description is that an ohmic contact is obtained when the interface energetics determine a larger carrier concentration at the interface than in the bulk at thermal equilibrium. In this case, it can supply current with a relatively negligible voltage drop since the interface is more conductive than the bulk. To give a microscopic description, it must be recalled that at such an interface E_F equalization at thermal equilibrium establishes a dynamic equilibrium: the diffusion of electrons from the metal to the semiconductor gives rise to a space charge in the latter, and the resulting electric field tends to drive electrons back in the metal until diffusion current is cancelled out by drift current. Upon application of an external voltage making the semiconductor more positive than the metal, net carrier injection occurs by partial suppression of the drift component in favour of the diffusive one.[27] The larger the applied voltage, the farther the system is driven away from thermal equilibrium. Roughly speaking, when the applied voltage completely suppresses the drift component, the contact is supplying the maximum current it is capable of, which is the diffusion current evaluated at the metal/semiconductor boundary. The larger is the difference between E_F and LUMO at the interface, the larger the maximum diffusion current and hence the larger the maximum current the contact is capable of. Once the current saturates, the contact is obviously no longer ohmic and the device becomes injection limited. Indeed, the maximum current is voltage dependent, since the Schottky effect[28] results in a lowering of the interface energetic barrier. In fact the superposition of the externally applied voltage with the image charge potential results in a local maximum for electron energy whose amplitude lowers when the applied voltage is higher (see **Figure 5**). This enhances the probability of metal to semiconductor injection thus extending the voltage range where the contact acts as ohmic.

The opposite situation occurs when the contact energetics determines a lower carrier concentration at the interface than in the bulk, thus giving rise to a non-ohmic contact and to an injection-limited current. In this case also, Schottky barrier lowering must be taken into account: a non-ohmic contact at low bias voltage might enter an ohmic regime from a certain threshold voltage onwards.

2.2.2. Injection Physics in Organic Semiconductors

Now we go beyond the rough classification of contacts into ohmic and non-ohmic, and turn to a more quantitative assessment of injection physics and of material parameters influencing it. We assume that E_F lies below the LUMO level by an amount Φ_B, the *nominal* energetic barrier (often referred to as Schottky barrier), so that to a first degree of approximation, negligible carrier concentration is expected in the semiconductor at thermal equilibrium.

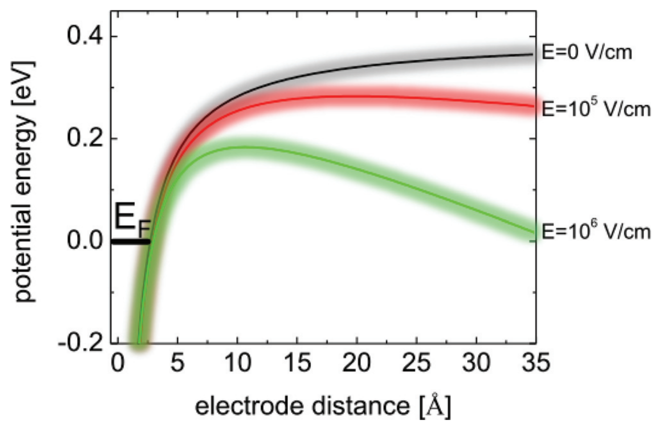

Figure 5. Potential energy distribution at the metal-semiconductor interface for three values of the externally applied electric field (nominal barrier Φ_B = 0.4 eV). Solid lines show the average electrostatic LUMO energy due to the superposition of the image charge Coulombic field and of the external electric field. The larger the external electric filed, the lower the maximum of the potential energy becomes (Schottky effect). The glow around solid lines qualitatively reminds that states are statistically distributed due to energetic disorder. Redrawn after Ref. [31].

For the description of carrier injection it must be considered that: *i*) transport in the disordered medium occurs by hopping, which is a thermally activated site-to-site tunnelling; *ii*) the superposition of the externally applied voltage with the image charge potential results in a local maximum for electron energy (Schottky effect); *iii*) the LUMO level represents only an average value of the energy of localized states, which are statistically distributed (supposedly according to a Gaussian) due to energetic disorder (Figure 5 summarizes these aspects).

First, we analyse the effects due to the first two aspects, and disregard disorder. For a metal electron having energy equal to or greater than the energy of unoccupied states in the semiconductor, injection can occur.[29] Due to hopping transport, electron excess energy is rapidly lost in the disordered medium and the thermalization length[30] is so short that only hops to the very first layers are to be taken into account (or equivalently long-range tunnelling plays no role[31]). Since typical jump distances are less than 1 nm, and since the potential maximum is comprised between 0.8 nm and 3.2 nm for electric fields ranging from 10^6 to 10^5 V/cm, the metal to semiconductor hop typically ends at the attractive part of the image potential. The fate of the excess electron can be either the drift back into the metal, which can be seen as Langevin bimolecular recombination with its metal image charge,[32] or the diffusion toward the semiconductor bulk. It is worth stressing that the low mobility of organic semiconductors requires the attractive part of the image potential to be taken into account, in contrast to high mobility crystalline semiconductors which thanks to their large mean free path easily overcome the potential maximum.

At thermal equilibrium injection and backflow cancel out, but upon application of external voltage, the energy maximum gets lower and closer to the interface, thus favoring injection with respect to backflow in the framework of a field-enhanced diffusion process. In the model proposed by Scott and Malliaras[32] (which disregards the effect of disorder), the injected current turns out to be equal to $J = e\mu E N_0 \exp(-\Phi_B/kT + \beta_s E^{0.5})\Psi(E)$, where μ is the carrier mobility, E the electric field, N_0 is the total site density, $\beta_s = [q^3 / 4\pi\varepsilon(kT)^2]^{0.5}$ embodies the Schottky effect, and $\Psi(E)$ is a correcting factor to be discussed below. In the following we briefly analyze the functional depedences of the model. The injection current has an Arrhenius-like dependence on Φ_B, due to the decreasing number of metal electrons with enough energy to be injected for higher (lower) barrier height (temperatures), just as in the standard thermoionic emission model.[28] Most notably, and unlike the standard thermoionic emission model, the injection current depends linearly on the carrier mobility, a fact that has been clearly proved experimentally.[33] This is due to the fact that carrier backflow obviously depends on mobility, but by detailed balance at thermal equilibrium, the same must hold true for carrier injection. This finding is worth stressing: the properties of a contact critically depend on the semiconductor mobility, thus the higher the mobility, the higher the net injection efficiency. Concomitantly, it must be recalled that if we assume for simplicity that the mobility of contact and bulk are the same, a higher mobility also implies a higher carrier transport capability of the bulk, so that injection issues will not be solved by simply enhancing carrier mobility.

The dependence on the electric field is dominated by a classical Schottky effect: the field induced barrier lowering $\Delta\Phi_B = (q^3 E/4\pi\varepsilon)^{0.5}$ results in an exponential dependence on the square root of the field, viz. $\ln(J) \propto \beta_s E^{0.5}$. But with respect to a classical Schottky effect, the absolute value of current is weakened by the Ψ term: this is due to carrier backflow, which is effective also at high electric fields. Finally it must be noticed that the dependencies highlighted so far on field and temperature are only a partial account of the dependencies of the injection current, because the mobility itself depends on field and temperature.

In the following we discuss the effect of energetic disorder. The statistical distribution of LUMO levels gives rise to a statistical distribution of energetic barriers. Hence Φ_B is to be understood as a nominal, *average* barrier, because even if the metal E_F lies below the nominal LUMO, it will lie above a portion of semiconductor tail states. This modifies the simple Arrhenius activation of current injection, weakening the temperature dependence.[31] In fact, at thermal equilibrium, at low carrier density and at relatively high temperature so that the Fermi-Dirac distribution can be approximated with the Boltzmann one, the *occupied* density of states has a Gaussian distribution centered at $-\sigma^2/kT$ with respect to the center of the DOS. The lower the temperature is, the farther this distribution lies from the nominal LUMO, thus making easier for a metal electron to jump into an unoccupied semiconductor state (**Figure 6**). This effect partially counterbalances the reduction in the number of hot electrons available at low temperature, and can be modeled in terms of a T-dependent effective barrier lowering $\Delta\Phi_B = -\sigma^2/2kT$.[34,35] For low enough temperatures, specifically when $T < \sigma^2/k\Phi_B$, the barrier lowering actually cancels out the nominal barrier, Fermi-Dirac statistics has to be fully taken into account, and we enter into a regime where

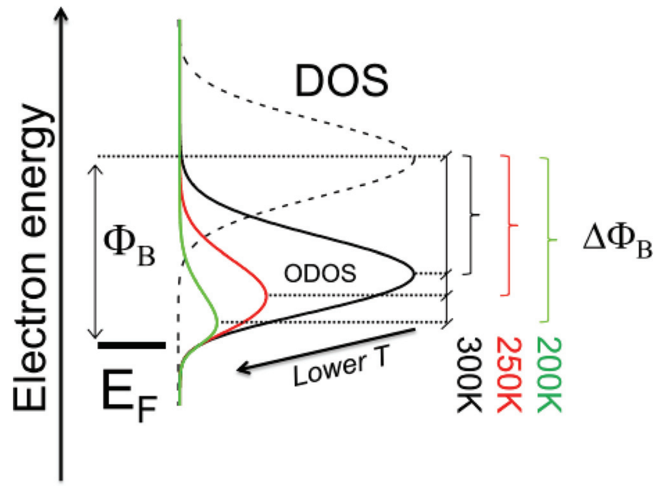

Figure 6. Effect of disorder on energetic barrier at the metal/organic interface. The nominal barrier Φ_B, which is the distance between the center of the DOS (σ = 100 meV, normalized plot) and the metal Fermi level, is 400 meV. At room temperature the density of occupied states (ODOS, black curve) closely resembles a Gaussian, and its center lies close to the metal Fermi level: the effective energetic barrier is lower than the nominal one by $\Delta\Phi_B = -\sigma^2/2kT = -123$ mV. Upon lowering the temperature (T = 250 K, T = 200 K) the barrier reduction is even larger.

the density of carriers at the interface is almost temperature independent, and remains finite even at 0 K (for analytic approximations holding in the low temperature regime see .. [35]). Since disorder induced barrier lowering is larger for more disordered systems, it might seem that disorder actually enhances injection, by populating interface tail states which can act as a carrier reservoir, but it must be taken into account that disorder negatively affects mobility as well. The final balance of these two effects is difficult to predict: Monte Carlo simulations[31] suggest that they cancel out for moderate barriers ($\Phi_B \leq 0.4$ eV), whereas a disorder-driven injection enhancement occurs for very large barriers ($\Phi_B = 0.7$ eV). Additionally, it must be noted that the energetic disorder at the interface is likely to be different from the one in the bulk: on the one hand, the presence of the metal boundary imposes an equipotential plane, thus largely suppressing electrostatic disorder for the very first layers;[36] on the other hand, it must be recalled from Section 2.1.1 that interfaces often show the presence of surface dipoles, which might be disordered, thus making the interface more disordered than the bulk.[37] In the latter case, the injection bottleneck could be represented not by the metal to semiconductor injection, but by the escape from the interfacial disordered region toward the bulk.

Finally, disorder affects the voltage dependence of injection current. The Schottky barrier lowering effect has to be modified to take into account the statistical distribution of barriers. Calculations supported by experimental evidence suggest that $\ln(J) \propto [(\beta_s+\beta_\sigma)E^{0.5}]$, with $\beta_\sigma = 2^{3/2}3^{-1}[qa\sigma^2/(kT)^3]^{0.5}$, where a is the intersite distance.[38]

2.3. Metal/Semiconductor Contacts in OFETs

We now focus on the injection physics in an OFET. This problem attains a higher degree of complexity, due to the fact

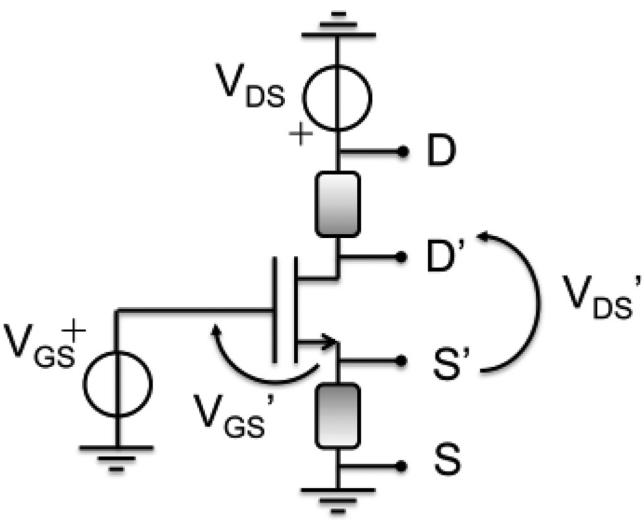

Figure 7. Lumped element model of a transistor with charge injection (and extraction) limitations. The externally applied voltages V_{GS} and V_{DS} do not drop entirely on the actual transistor; the voltage drops driving the transistor are V_{DS}' and V_{GS}'. The shaded two-terminal boxes in between the physically accessible terminals, S and D, and the physically inaccessible transistor terminals, S' and D', represent non-ideal contacts.

that it is two-dimensional in nature, and that the electrostatic action of the gate can modify the source/drain injection properties. As a simplifying point of view, a *lumped element model* is often adopted, and a real OFET, characterized by non-ohmic contacts, is modeled as an ideal OFET having two-terminal elements in series to source and drain representing the effect of the contacts. We call the physically accessible terminals Drain (D) and Source (S), whereas we call D' and S' the inaccessible terminals of the ideal OFET, as schematically shown in **Figure 7**. The externally applied voltage V_{DS} (V_{GS}) partly drops across the ideal device giving rise to V_{DS}' (V_{GS}') and partly drops across the contacts. We will collectively call the voltage drop across the contacts $V_C = V_{DS} - V_{DS}'$. In the so-called linear region of OFET operation ($V_{DS} << V_{GS} - V_T$, where V_T represents the threshold voltage[39]), which will be assumed in the following unless otherwise stated, it is usual to approximate V_{GS}' with V_{GS}, since $V_C << V_{GS} - V_T$. Therefore the net effect of non-ohmic contacts is to diminish the voltage drop across the channel. If the current-voltage characteristic of contacts is linear, we will talk of *contact resistances* (R_S and R_D in series to source drain respectively, $R_C = R_S + R_D$), otherwise non-linear injection takes place. As a first approximation, the channel is assumed to be undisturbed at least far away from the semiconductor/contacts boundary.[40,41] On the basis of the comparison between the charge density at the semiconductor/contacts boundary and the one in the channel due to the gate action, the contact region is said to be "depleted" (with respect to the channel) and it is expected to yield a poorly injecting contact, or conversely, the contact region is said to be "enriched", and nearly ohmic behaviour is expected.

It is worth including a caveat at this point. The description given above relies on the hypothesis of thermal equilibrium: despite poorly injecting contacts, charge accumulation occurs and the channel is formed. Indeed, since channel carriers must be injected from the contacts (thermal generation is usually negligible), in case of very poorly injecting contacts, it is questionable whether the channel accumulation regime can be attained at all.[42] In case of very large barriers, it is doubtful whether device performances are poor because contacts impede current injection (and extraction) into (and from) an existing channel, or because the channel itself barely exist. As a final remark, static device simulations, which are performed assuming thermal equilibrium, should be considered with caution in case of large barriers.

2.3.1. OFET Topology and Injection

From a *topological* point of view, we distinguish between coplanar devices, where the transistor accumulated channel and the source/drain belong to the same plane and lie at the semiconductor/dielectric interface, and staggered devices, where the semiconductor is in between the dielectric layer and the plane containing source and drain. Further nomenclature follows on the basis of the processing scheme, as shown in **Figure 8**. Coplanar devices can be bottom-gate, bottom-contacts (BGBC), with the semiconductor deposited as the last step, or top-gate, top-contacts, with the metal gate deposited as the last step and source/drain deposited on the semiconductor. In the case of solution processed semiconductors the latter architecture is not actually adopted. Staggered architectures can be top-gate, bottom-contacts (TGBC), with source/drain deposited in the first step and metal gate in the last one, or bottom-gate, top-contacts (BGTC), with gate contact deposited in the first step and source/drain in the last one. Processing effects on injection will be dealt with in Section 2.5.

There is large consensus both from a theoretical[41,43–47] and experimental[40,44,48] point of view that staggered configuration yields better contacts, for a given metal/semiconductor choice. To a first degree of approximation, this can be understood in terms of geometrical reasons: for the coplanar configuration injection occurs from the contact edge, whose thickness is usually few tens of nm, into the accumulated channel, whose thickness is a few nm; for the staggered configuration in principle all the contact area facing the gate contact can be involved in the injection process.[45] In the following the two configurations are analysed separately, and the statements above will be detailed and refined.

2.3.2. Coplanar OFET

In the case of coplanar topology few examples of enriched contact regions can be found in the literature: for instance poly(3-hexylthiophene) (P3HT) with Pt contacts, or again P3HT with Au contacts (but in this case it is suggested the excess carriers do not arise from a metal to semiconductor charge transfer but from unintentional doping).[49] In the following we focus on the physics of depleted contacts.

The depleted region forms in between the metal contact edge and the accumulated channel, thus giving rise to a bottleneck to carrier injection and extraction. To a first degree of approximation, current flows from and to the contact edges, hence contact thickness plays a role[45] and of course very thin contacts are detrimental. Indeed, the role of top surfaces in carrier injection

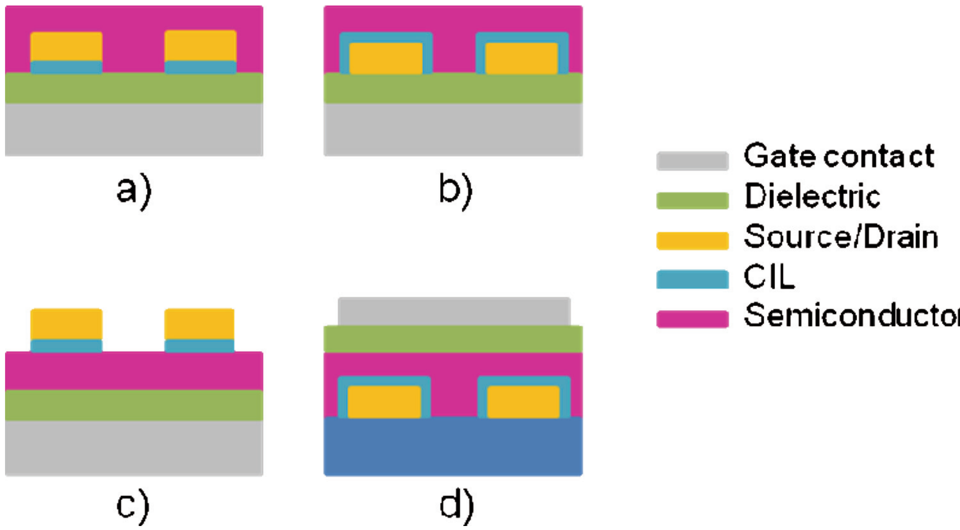

Figure 8. Schematic sections of the most widely adopted architectures for OFETs; charge injection layers (CILs) are indicated as well and can either be present or not. a) and b) BGBC structure; here an adhesion layer can serve as CIL (a) or this can be formed on top of the electrodes (b). c) BGTC device, where a CIL deposited on the semiconductor prior to the electrodes growth. d) TGBC structure, with a CIL similar to case b.

has been experimentally shown to become relevant in case of very thin (below 15 nm) contacts[50] (possibly current crowding effects occur in this case,[51,52] see Section 2.3.3).

We start with an ideal situation, where the semiconductor active layer is trap-free and uniform from source to drain. The situation at source and drain is not symmetric: larger voltage drops are expected at the source, where carriers move from a depleted towards an accumulated region. Indeed, simulations show that for moderate energetic barriers ($\Phi_B \leq 0.3$ eV), voltage drops are negligible, and become sizeable only for larger barriers and only at the source side.[45,53,54] For $\Phi_B \leq 0.3$ eV, the lateral extension of the depleted region depends on the interplay between the height of the energetic barrier and the channel screening length, which in its turn is a decreasing function of the gate voltage (at low gate voltage, when the screening length is very long, the depleted region is upper bounded by the film thickness).[41,55]

Now we turn to a more realistic scenario, and we take into account that close to the contact the active layer morphology is often different and less favourable to carrier transport than it is far from the contact, due to the existence of the triple interface between the contact, the dielectric and the semiconductor. When low-mobility (or strongly trap-depleted) regions exist adjacent to the contacts, simulations show that injection predominantly occurs from the contact top surfaces, and these conduction paths give rise to sizeable, symmetric at source and drain, contact resistances even for moderate barriers ($\Phi_B \leq$ 0.3 eV).[53] It has to be noted that if the semiconductor suffers from low mobility all around the contact, and not only close to the edge, injection returns to be localized at the contact edge.[44]

Long conduction paths due to top surface injection are not the only possible explanation for contact resistances in case of moderated barriers. In the work by Koehler et al.[34] contact resistances are modelled as due to trapping of gate induced charge and to space charge limited conduction close to the contact (injection is modelled according to the Scott-Malliaras model modified to take disorder into account).

In all the cases above, injection limitations can be modelled as contact resistances, which are increasing functions of Φ_B, and decreasing functions of carrier mobility and gate voltage. This scenario is in agreement with experimental results obtained by means of direct techniques such as Scanning Kelvin Probe Microscopy[56] (SKPM) or Gated Four Point Probe[57] (gFPP, see Section 3.2). An upper limit to the extension of contact regions in the range of 100 nm can be deduced from SKPM and gFPP measurements.

In some situations, injection limitations cannot be modelled in the form of contact resistances due to their **non-linearity**. This occurs when: *i*) large energetic barriers are present, for instance in case of P3HT with Cr contacts, where Φ_B is expected to be larger than 0.3 eV[58–60]); *ii*) accessible site density in the semiconductor is lower than the monomer density[61] and *iii*) in case of low barriers, but taking into account the charge and field dependence of the mobility along the depleted contact region. The latter effect requires a strong mobility field dependence, such as the one found in the Pasveer/Coehoorn model[62] where $\ln(\mu) \propto E$ for high fields, while it is barely noticeable if the mobility has a Poole-Frenkel dependence, $\ln(\mu) \propto E^{0.5}$.[47] It has to be noted that non-linear injection can alter threshold voltage extraction if not properly taken into account.[61,63]

Regarding the temperature dependence of injection, again two regimes exist on the basis of a threshold value for Φ_B around 0.3 eV. For $\Phi_B \leq 0.3$ eV injection currents have an Arrhenius-like behaviour, with an activation energy closely resembling the mobility activation energy. Recalling that injection current is proportional to the mobility, we conclude that the latter actually dominates the injection T-dependence (at least down to 140 K).[64] It is likely that the negligible contribution from Φ_B to the temperature dependence can be attributed to the disorder induced barrier lowering ($\Delta\Phi_B = -\sigma^2/2kT$). For barriers larger than 0.3 eV, experimental data show activation energies lower than both the mobility activation energy and Φ_B.[40,56,64] In this case, in addition to the aforementioned disorder induced barrier lowering, the disorder-enhanced Schottky effect ($\Delta\Phi_B \propto E^{0.5}$) is likely to become very effective, since in the case of large barriers high electric fields at contacts are expected. In addition, for very low temperatures, $T < \sigma^2/k\Phi_B$, it must be recalled that barrier vanishes and a regime of negligible T-dependence is expected (for instance in case of P3HT with Cr contact this threshold temperature value is evaluated to be at 160 K).

The practical consequence of the weaker T-dependence of injection current with respect to mobility, is that injecting properties improve *relative* to the channel as temperature is lowered; hence a device that is not contact limited at room temperature will remain so at lower temperatures.[64]

To summarize the coplanar configuration, we can say that carriers are injected into the accumulated channel laterally: roughly speaking this limits the order of magnitude of the injection area to the channel depth, which is in the range of few nanometers.[65] Either injection occurs in a very effective way, or this represents a bottleneck. Injection from the top surface, which might take advantage of a larger area, is normally inconvenient because the current would flow through highly resistive (not modulated by the gate) regions, and becomes convenient only when edge regions are characterized by low mobility.

2.3.3. Staggered OFET

In the following we assume that a large overlapping of the source and drain with the gate ensures that the channel extends far beyond the projected source to drain gap. Then, in the bottom(top) gate configuration, carriers are injected from the bottom(top) surface of contacts, which can be large, travel across the semiconductor film thickness, reach the accumulated channel and drift along it. Contacts with a low current injection efficiency per unit area (or paths from contact to accumulated channel with high resistivity), can be, at least partially, compensated for by extending the involved area farther from the projected source to drain gap. Thanks to this self-regulating mechanism, the sensitivity of the contacts to the barrier height is milder than in the coplanar configuration.[40,41,45,47,66]

The semiconductor thickness has to be judiciously chosen, and an optimum is expected to exist: if too thin, the staggered configuration fades into the coplanar one (and from an experimental point of view very thin layers can be difficult to obtain or can show a morphology unfavorable to transport);[40,67] if too thick, the transport from the metal contacts to the accumulated channel may result in a bottleneck:[67–70] this can be counterbalanced by involving larger injecting areas, but once the entire contact area is employed, the mechanism ceases to be effective. On the other hand, the contact thickness is expected to play a minor role.[45]

The advantages of the coplanar configuration are obtained at the expense of: *i*) the occupied device area (the involved injecting contact length is estimated to range from few micrometers up to tens of micrometers);[68,71] *ii*) the device frequency response, because the source and drain to gate overlaps result in relatively large parasitic capacitances.[72] Regarding the latter issue, it has to be noticed that when the device is biased in saturation, no accumulation channel exists on the drain side, hence current flows into the drain contact only through its edge and not through its top or bottom surface. Therefore the dielectric of the gate to drain parasitic capacitance is a double layer stack comprised of the semiconductor and of the OFET gate insulator. This has to be compared to the coplanar configuration, where the gate to drain overlap capacitance is given by the sole thickness of the OFET gate insulator (regardless of the biasing in linear or saturation regime). Simulations suggest that the gain-bandwidth product of a common source staggered OFET can be maximized through proper engineering of the semiconductor thickness (parasitic capacitance and gain, which are proportional to the transconductance, are respectively a decreasing and an increasing function of the semiconductor thickness), and can exceed the gain-bandwidth product of a coplanar common source OFET.[70]

Even in the case of negligible gate to source/drain overlap, which would be beneficial for the device frequency behavior, the staggered configuration surpasses the coplanar one in terms of injection efficiency, thanks to a certain spreading of the gate effect, which creates accumulated regions extending beyond the projection of the gate onto the semiconductor/dielectric interface plane.[47]

In the following we discuss the mechanism regulating the contact injecting area, also known in the literature as *current crowding effect*. To illustrate the physics behind this phenomenon, we start with a very simple and idealized model:[71] *i*) gate-contacts overlaps extend indefinitely; *ii*) mobility is constant; *iii*) injection and transport across the semiconductor are modeled as purely resistive; *iv*) current flows from the contact to the accumulated channel parallel to the y axis, and then the flow proceeds along the accumulated channel parallel to x-axis (**Figure 9**). We refer here to a p-type OFET. According to Figure 9, $J_y(x)$ is the current per unit surface flowing across the semiconductor, the latter being characterized by a resistance per unit surface $R_y = R_{contact} + R_{bulk}$; $I_x(x)$ is the current flowing along the accumulated channel, the latter being characterized by a sheet resistance assumed to be constant along x and equal to $R_{sh} = [\mu C_{ox}(V_{GS} - V_T)]^{-1}$. The current $I_x(x)$, for $x < 0$, is given by the integral sum of the infinitesimal contributions $\int_{-\infty}^{x} J_y W dx'$ in the region $x' < x$. Therefore, $I_x(x)$ is not constant but increases for x approaching $x = 0$, where it reaches its maximum, and it is constant for $0 < x < L$. Since $I_x(x)$ is

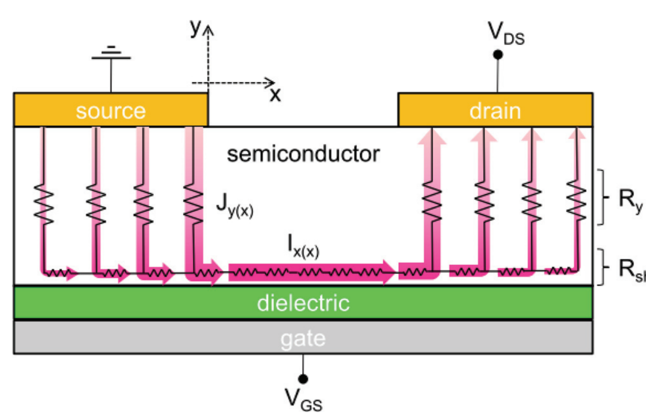

Figure 9. A sketch of a TCBG transistor affected by current crowding effect. Under source (drain) contact, $J_y(x)$ increases (decreases) moving towards (away from) the projected source-draing gap edge. In the gap region current flows only along x axis. Redrawn after Ref. [68].

an increasing function of x for $x < 0$, but R_{sh} is constant, the potential along the channel $V(x)$ has to decrease superlinearly with space. The final consequence is that regions closer to $x = 0$ are more strongly biased, and the current density $I_y(x) = |V(x)|/R_y$ is an increasing function of x, crowding close to the source/drain gap. From a more quantitative point of view, by solving the problem in the framework of the transmission line model, it is found that: *i*) the x-dependence of J_y, I_x, and V_x is exponential with a characteristic length $L_O = (R_y/R_{sh})^{0.5}$; *ii*) the source contact resistance R_S, calculated as $|V(0)|/ I_x(0)$, can be expressed as $R_S = R_y/ (WL_O) = (R_yR_{sh})^{0.5}/W$, the total contact resistance R_C being twice this value (under the simplifying assumption that R_y is the same for source and drain[73]).

The basic features which emerge are: *i*) R_C depends on injection/transport along y-axis (through R_y) and on transport along x-axis (through R_{sh}), since the accumulated channel in the overlapping region is part of the access path to the channel (quantitatively speaking R_C scales as the reciprocal of the geometrical mean of the mobilities along y and along x); *ii*) since R_{sh} is a function of the gate voltage, R_C inherits a gate voltage dependence, even if injection in itself is not gate voltage dependent; *iii*) the dependence on R_y introduces a dependence for R_C on the semiconductor thickness. In addition, if we assume for simplicity that both R_y and R_{sh} have an Arrhenius-like dependence on temperature, the activation energy of R_C will be given by the arithmetical mean of the activation energy of R_y and R_{sh}. If we further assume that these activation energies are similar, then R_C and R_{sh} share the same dependence upon temperature, hence a device which is not contact limited at room temperature will remain so at lower temperatures.

The physical behavior of the current crowding effect is the following: if R_{sh} gets lower, either because of larger gate voltage or of higher mobility, the injecting area is enlarged as a reaction to the augmented current capability of the channel so that R_C is also lowered ($R_C \propto R_{sh}^{0.5}$). If we define the ratio R_C/R_{sh} as a figure of merit for the relative weight of contact resistance, this quantity turns out to be proportional to $1/R_{sh}^{0.5}$. We now compare this to the case of constant R_C, which would yield the R_C/R_{sh} ratio to be simply proportional to $1/R_{sh}$. While it is clear that for negligible R_{sh} in both situations the R_C/R_{sh} ratio diverges and the device becomes contact-limited with V_{DS} dropping entirely with R_C, the staggered configuration advantageously approaches this limit more slowly, thanks to the square root dependence on $1/R_{sh}$. While approaching the contact-limited regime, in a device with constant R_C, the current tends to become increasingly independent on V_{GS}, whereas in the staggered configuration, since R_C is V_{GS}-dependent, the current is still modulated by the gate, albeit with a sub-linear square-root functional dependence on V_{GS}.

We now come to more detailed models for current crowding. If the simplifying assumption of infinite contact/gate overlap is removed in favor of a finite overlap equal to L_{OV}, an upper (lower) bound to the injecting area (R_C) is introduced.[74] When L_O approaches L_{OV}, R_C becomes constant and equal to $2R_y/(WL_{OV})$.

Resistive transport along the y-axis is not expected to be a proper model for organic layers (if we exclude doped layers), and the community has focused on refining modeling in order to improve the fitting of experimental data. In the literature two approaches emerge. A first group of authors[67,69] focuses on the fact that the gate action is not entirely confined at the dielectric interface, but it extends deeper, albeit mildly, into the semiconductor, thus making transport along the y-direction V_{GS}-dependent and finally resulting in a R_C dependence on V_{GS} close to linear. In addition, ref.[69] accounts for injection limitation by means of a non-zero voltage boundary condition at the metal/semiconductor interface: it is shown that the V_{GS} dependence of the contact resistance is larger when injection limitations occur.

A second group of authors describes the y-transport in the framework of space charge limited conduction (SCLC) with traps[52,70] or in a trap-free scenario,[68] thus making the excess carrier density along y-axis governed by the voltage drop on the semiconductor thickness rather than by the gate voltage. The peculiar super-linear power-law dependence of current on voltage and thickness in SCLC modifies quantitatively the R_C dependence on thickness and gate voltage already described for the basic model and most notably, causes R_C to become dependent on V_{DS} (the effect being more evident at moderate V_{GS}). A major consequence of the latter phenomenon is that the total device resistance is not a straight line as a function of the channel length L, but proves to be slightly convex (this fact makes the application of the transfer-line method questionable, see Section 3.1.1).[68,75]

There are other situations which can lead to non-linear injection, namely: *i*) large mobility anisotropy, with the mobility across the semiconductor being 3–4 orders of magnitude lower than the mobility along the accumulated channel;[42,46] *ii*) large distribution of traps (in the range of 10^{19} cm^{-3});[42,46] *iii*) parasitic currents charging the device periphery and preventing the formation of the channel at the semiconductor/dielectric interface under the source and drain contacts.[76] Non-linear behaviour due to large Φ_B might be negligible due to the self-regulation of the involved injecting area.[66]

2.4. Charge Injection Control Through Contact Modification

In order to reduce contact resistance effects in OFETs and allow scalability of the channel length, various approaches have been developed and tested in the literature so far. Despite a general approach for the realization of ohmic contacts not yet having been developed and the fact that most of the time the techniques are material or architecture dependent, some of them are quite well established and serve at least as a starting point for the optimization of charge injection. In the following we will review, thanks to notable examples, the most widely adopted approaches, such as the matching of the electrode W_f to the frontier molecular energy levels of the semiconductor by adopting different conductors, or by engineering the effective W_f by means of self-assembled monolayers (SAMs) or other charge injection layers (CILs). The aim is to provide the reader with an overview on the most effective and documented strategies for contact resistance reduction, while keeping in mind that at the metal/organic semiconductor interface many and interacting effects take place at the same time, as described in Section 2.1, and it is not always simple to disentangle them.

REVIEW

2.4.1. Energy Level Matching by Variation of the Metal Electrode

The finite Schottky barrier at the metal-semiconductor interface in an OFET can be nominally tuned by adopting a conductor characterized by a suitable W_f. Given that most of the organic semiconductors adopted so far in OFETs have an energy bandgap $E_G > 2$ eV, with a LUMO level lying between −2 and −4 eV, and a HOMO level between -5 and -6 eV, specific electrodes have to be adopted for the specific charge carrier whose injection has to be optimized. Therefore, as a rule of thumb high W_f electrodes have to be adopted for hole injection, while low W_f electrodes are preferable for electrons. Although this approach is simplistic, it is possible, at least to some extent, to find in the literature a few examples showing that height of the Schottky interfacial barrier can be effectively tuned in this way. For hole injection clear dependence of the contact resistance on the electrode W_f was observed both for P3HT and poly-dioctylfluorene-*co*-bithiophene (F8T2) BGBC devices.[56] In the first case values as low as a few kΩ·cm were reported in the case of Au contacts, while a few MΩ·cm are obtained when lower W_f electrodes such as Cr are employed.

High W_f metals such as Ca were instead crucial to demonstrate *n*-channel, and more in general ambipolar transport in OFETs.[77] Ca or Au capped Ca electrodes allowed to achieve very low R_C values of a few kΩ·cm, a very remarkable value for *n*-channel OFETs, in a BGTC [6,6]-phenyl C61 butyric acid methyl ester (PCBM) device.[78]

With the recent higher availability of small bandgap polymers, synthesized with the aim to enhance ambipolarity, it has been shown that reasonably efficient electron injection can also be achieved from high W_f metals such as Au, most likely due to a combination of the reduced E_G and of the push-back effect. For example values ~20 kΩ·cm were recently achieved in a TGBC high-mobility *n*-channel polymeric OFET adopting Au electrodes.[75]

As already mentioned, it is well established now that alignment of the clean metal W_f to the semiconductor frontier orbitals offers no more than a first rough indication, since there are demonstrated effects that produce substantial energetic shifts at the interfaces. Importantly they are not always under control and strongly depend on the sample processing conditions. We recall here for example that injection from Au contacts in P3HT is favored in BGBC OFETs by dopants normally present in P3HT exposed to ambient conditions, and contact resistances in these devices can be drastically affected by dedoping through an annealing process, while less severe effects are observed with Pt electrodes.[49] Moreover, we previously saw that the effective W_f of a metal can shift quite substantially even in the absence of surface dipoles, due to the push-back effect, as a result of the compression exerted by the organic materials on the metal electron-density tail. On Au surfaces this effect can account for significant shifts up to ~1 eV.[17,18] Therefore untreated Au electrodes can be a far worse hole injector than nominally expected, and conversely can work as a good electron injector for micrometer scale channel lengths in some low band-gap polymers.[75]

Keeping this is mind, it is clear that the selection of a suitable electrode based on the minimization of the energetic mismatch between the frontier molecular orbitals and E_F is common practice in research. This is meaningful especially in the case of high band-gap materials, as shifts are usually lower than 1 eV. It must however be underlined that with the recent development of a wide range of low band-gap materials,[79–83] these effects can quite substantially and unexpectedly affect contact resistances.

2.4.2. Control of the Interfacial Dipoles Through Thiol-Based Self-Assembled Monolayers

Reduction of contact resistance effects through the variation of the metal electrode is clearly not a technologically appealing approach as real processes are usually compatible with a limited set of conductors. Therefore a more general approach is required. Moreover in *n*-channel FETs low W_f electrodes are usually required, with the obvious drawback of poor device stability due to their strong reactivity.

One of the most successful approaches for the control of metal/organic semiconductor interface Schottky barriers is the use of polar, thiol-based self-assembled monolayers. Ordered SAMs can be obtained by dipping the electrode in 1 to 10 mM solutions of the desired thiols. An alternative approach is to deposit the molecules from the vapour phase in a saturated atmosphere. Thiols bind to the metal surface, preferably gold and silver, thanks to the sulfhydryl group. This occurs through a chemisorption process, where the sulphur atom forms a covalent bond with a metal atom on the surface and the hydrogen atom is released. The first practical use of this approach for the control of interfacial energy barriers was demonstrated for vertical diodes in the early work of Campbell et al.[84] Molecules with an intrinsic electric dipole moment ρ (**Figure 10**a), which depends on the particular chemical structure, can induce a shift in the surface W_f proportional to ρ. W_f can effectively be increased or decreased by up to 1 eV depending on the orientation of ρ with respect to the surface (Figure 10b). Ideally W_f should be modulated in order to minimize the energy mismatch with either HOMO or LUMO, depending on the injected carrier. It is not however advisable to stop at these boundary levels. In fact, as a consequence of what was described in Section 2.1.1, the lower (higher) the E_F is with respect to the HOMO (LUMO) level, the higher the charge transferred to reach thermal equilibrium and the lower R_C is.

Despite some initial controversy about the extension of this approach to OFETs,[85] a clear correlation between R_C and W_f induced by different thiols has been observed in the literature,[86,87] demonstrating the effectiveness of this approach in reducing contact resistance effects in OFETs, both in *p*- and *n*-channel solution processed OFETs.[87–92]

Importantly, the growth of SAMs on electrodes can quite dramatically alter their surface energy, thereby also inducing other effects superimposed on the variation of the W_f, making interpretation of the device characteristics challenging. For example a different degree of wettability was found to induce a different thickness of a poly(9,9-di-n-octylfluorene-alt-benzothiadiazole) (F8BT) thin film spun on top of Au electrodes treated with 1-Decanethiol (1D), a common thiol adopted to improve electrons injection. A lower degree of wettability produced a thinner F8BT film, thus enhancing injection of both holes and electrons in top-gated ambipolar OFETs.[89] Indeed film thickness variations generally have a strong effect on R_C in the case of staggered architectures (see Section 2.3.3) since

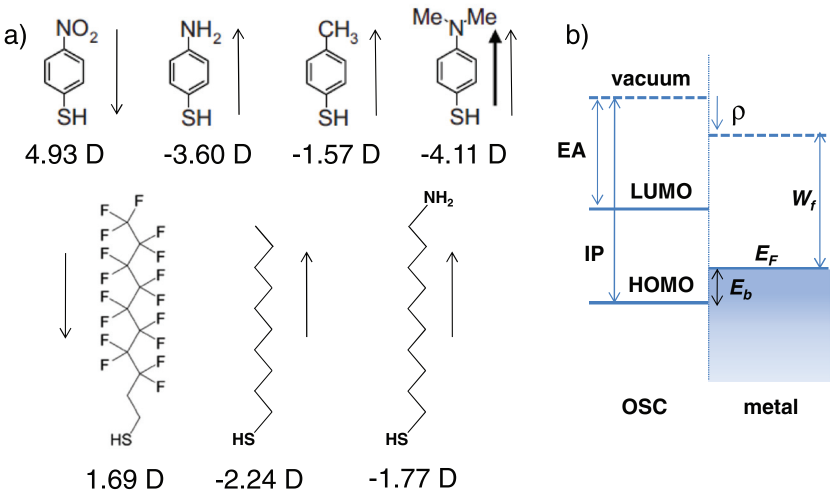

Figure 10. a) Some of the most widely adopted thiols for the modification of metal contacts in OFETs with indication of dipole moments represented in debye units D; here an arrow facing upward represent a dipole that facilitates electrons injection. Adapted with permission.[221,89] Copyright 2009, American Institute of Physics and WILEY-VCH Verlag. b) Schematic energy levels diagram at the interface evidencing the effect of the thiol dipole moment ρ.

they affect the access resistance R_γ to the accumulated channel through the bulk region.[67,75,86]

Another important aspect to take into account is that a variation in the wettability induced by the SAM growth can also have an effect on the film morphology at the interface with the electrode (**Figure 11**).[56,93,94] Although a general understanding has not yet been achieved, it is quite established that control of the morphology at the contact edges is critical for efficient charge injection in BC devices. Since the semiconductor molecules may experience a variation in surface energy from the electrode area to the substrate area, it is possible that the morphology and degree of order are affected. This is clearly observed for example in BG evaporated pentacene devices, where the molecules have an edge-on orientation on silicon-dioxide, whilst laying flat on gold electrodes. This leads to a strong contact resistance due to a wide transition area at the edge of the contact, where the order is disrupted.[95] A similar effect was observed in solution-processed poly(2,5-bis(3-tetradecylthiophene-2-yl)thieno[3,2-b]thiophene) (pBTTT) top-gate devices, where injection occurs in a current crowding regime (see Section 2.3.3) and hence the access resistance R_γ to the accumulated channel plays a key role in determinig R_C. The functionalization of gold electrodes with a perfluorinated 1H, 1H, 2H, 2H-perflourodecanethiol (PFDT) SAM and similar functionalization of the substrate with octadecyltrichlorosilane (OTS) allowed a uniform morphology to be obtained, whilst simultaneously lowering the contact W_f.[96]

It was reported that this effect appears not to be crucial in prevalently amorphous films,[89] while it is more important in polycrystalline semiconductors. In the latter case, on the one hand variations in thickness themselves can produce a different film morphology and surface roughness, important in TG OFETs, thus affecting the intrinsic carriers mobility.[86] On the other hand, correlations between the film grain sizes and the contact surface energies were observed for solution-processed polycrystalline thin films. Extended grain sizes favors charge transport as long as a good inter-grain connectivity is preserved, and can therefore lead to an increased

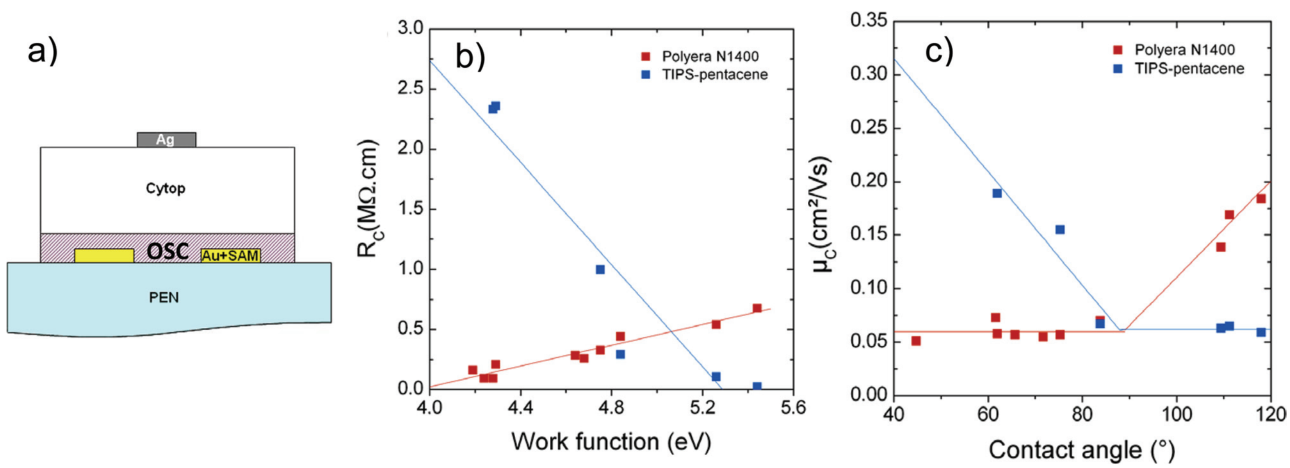

Figure 11. a) Schematic cross-section of a TGBC solution-processed OFET on a polyethylene naphthalate (PEN) substrate, where the Au electrodes are treated with a SAM to tune their W_f. b) Dependence of R_C on the electrode W_f for the device in panel a) adopting two distinct polycrystalline semiconductors. In the case of the Polyera N1400 *n*-type semiconductor, R_C can be reduced by reducing the electrode W_f to better match the LUMO level (4.3 eV) of the molecules. In the case of the *p*-type TIPS-p (HOMO level at 5.3 eV), the opposite is true, R_C is reduced when W_f is increased. c) Dependence of the field-effect mobility corrected for R_C on the contact angle with water for the device in panel a) adopting the same polycristalline semiconductors. In the case of *n*-type N1400 mobility is unaffected for an angle below 90°, while it linearly increases for more hydrophobic substrates. In the case of *p*-type TIPS-p an opposite behavior is found, with a constant mobility for angles above 90° and a linear dependence for more hydrophilic substrates. Adapted with permission.[88] Copyright 2010, Elsevier B.V.

intrinsic mobility. Since the semiconductor morphology at the contact alters both the carriers mobility and the interface energetics, and both mobility and energetics influence R_C, disentangling the two effects is not trivial.[88]

In the literature, the introduction of a tunneling barrier by the presence of a SAM has often been overlooked.[97,98] Therefore a contribution to the total R_C is expected to come from the tunneling resistance of the SAM molecule. This can compete with the effects previously described, depending on the actual tunneling length, electronic states of the SAM and which carrier is being injected.[89,97]

2.4.3. Charge Injection Layers in OFETs

In this section we will review the most widely known charge injection layers (CILs) adopted to reduce contact resistances in OFETs. Rigorously, the thiol-based SAM approach we treated in the Section 2.4.2 is a particular case of CILs, however given the generality of the former approach and the extensive literature, we dedicated a specific paragraph to it. Apart from thiols, other CILs have been adopted in OFETs, from physisorbed thin films to chemisorbed (sub)-monolayers. This is a common approach adopted in OLEDs[99,100] to balance electron and hole injection, something which indeed benefited the development of CILs for OFETs. We first review the approaches for hole and electron injection, and we then focus on the adoption of solution-processible CILs for contact engineering in complementary digital circuits.

In Figure 8 the most widely adopted architectures where CILs have been introduced are shown. In a BGBC OFET, the interlayer can replace the usual metal adhesion layer (Figure 8a), such as Cr or Ni. In this case it is found that a suitable CIL can be effective in enhancing the injection in the thin accumulated channel at the semiconductor/dielectric interface.[101] In BGBC OFETs it is more common to find CILs which are placed on top of metal electrodes (Figure 8b), where due to a variation in surface energy or to a higher interfacial interactions in the case of organic CILs, morphological effects in the semiconducting film can play a major role (see Section 2.3.2 on the relevance of top surface injection in coplanar OFETs). In a BGTC device (Figure 8c) the interlayer affects the device in multiple ways: similarly to the BGBC case it can help to reduce the effective barrier at the interface through favorable level alignment, but it additionally acts as a protective layer against the penetration of evaporated contacts in the soft semiconductor.[102] Apart from a protective role, the previous effect can also favor injection in TGBC devices (Figure 8d).

In *p*-channel devices it was found that it is possible to reduce R_C by inserting a thermally evaporated transition metal oxide interlayer such as MoO_3 or WO_3[103–105] between a thermally evaporated contact and the semiconductor. The effect was demonstrated in BGBC FETs with vacuum sublimed pentacene, leading to contact resistances of a few $k\Omega \cdot cm$[101] and BGTC FETs based on pentacene ($R_C \sim 1\ k\Omega \cdot cm$) and other small molecule semiconductors.[106] The origin of injection improvement was the subject of debate and recently it was demonstrated that a local enrichment of charge density in the semiconductor interface with the MoO_3 layer can occur due to interfacial *p*-type doping provided that the electron-affinity (EA) of the oxides is deeper than the HOMO level of the semiconductor;[107,108] concomitantly it was shown that the same approach should be applicable to solution-processed polymers, although exposure of devices with a MoO_3 interlayer to ambient conditions can cause severe degradation due to variations in the oxides energy levels.[108] The same interfacial mechanism is also suggested to be valid in the case of WO_3, while for other oxide interlayers a better level alignment is the suggested reason for enhanced injection, as in the case of Cu_xO in TC OFETs formed during deposition of Cu contacts due to the oxygen present in the semiconductor film even at high vacuum.[109,110] This is very beneficial as it changes inexpensive Cu from a poor to a good hole injector.

Interfacial doping, was also demonstrated in the case of pentacene by means of $FeCl_3$ both in staggered and coplanar OFETs,[111,112] but an open question remeains regarding the long-term stability of such an approach in a real device, where dopants could diffuse within the semiconductor under device operation.[111,113]

Another established approach to increasing the electrode W_f and reduce the injection barrier for holes in OFETs is based on the adoption of **strong acceptors**, even monolayers, such as the well known tetracyanoquinodimethane (TCNQ) and tetrafluorotetracyanoquinodimethane (F_4TCNQ).[114–116] This approach was first demonstrated for the treatment of Au contacts,[114] but its effectiveness was then assessed also for cheaper contacts such as Cu and Ag,[117,118] offering quite a general approach for the optimization of contacts in organic electronics.[119,120] The underlying mechanism was clearly attributed to a balance between forward and backward charge transfer between F_4TCNQ and the metal, and a geometrical modification of the adsorbed molecules on the metal.[118,121] The net effect is an increase in the metal W_f and therefore a lowering of the injection barrier. Such approach has also been demonstrated to be compatible with solution processing, since spin-coated TCNQ or F_4TCNQ solutions can form strongly bound monolayers on metal electrodes through chemisorption (TGBC OFET based on poly(3-hexylthiophene)-dithienyltetrafluorobenzene).[122]

While the mechanism which leads to W_f increase was clarified, allowing R_C reduction in pentacene and polythiophene derivative based devices, it was shown that the same treatment was also surprisingly beneficial for *n*-type thermally evaporated $F_{16}CuPc$ OFETs. This was explained by better compatibility of the modified contact with the vacuum deposited molecules, favoring the growth of larger crystals on the electrodes, comparable to the ones in the channel,[110,117] pointing out that morphological effects can also play a role in this case.

Similarly, a lower barrier and a strong π-π interaction at the interface between pentacene and graphene modified contacts, effectively reduces contact resistance for holes mainly for morphological reasons, e.g., formation of larger pentacene grains over the contacts. This approach also allows the adoption of low cost metal electrodes such as Ag or Cu, which would produce a large injection barrier for *p*-type FETs if used without a CIL.[110,123]

CILs have also been developed to enhance **electron injection**, although to a much lesser extent given the very recent development of high performance, stable *n*-channel OFETs.[124–127] *N*-type metal oxide CILs on metal electrodes can provide efficient electron injection. For example ZnO and InZnO covered

Au electrodes were shown to strongly reduce contact resistances in polymers with low EA thanks to a reduced energy barrier,[128] allowing better balancing of electron and hole injection in light-emitting field-effect transistors (LEFETs). These oxides CILs are solution processible, thanks to the use of precursors, and their processing temperatures can be lowered to be compatible with plastic substrates, thus representing a very promising option for the development of low-cost, solution-processed FETs.

Similar to the approach of coating metals with strong molecular acceptors in the case of hole injection, it was shown that neutral methyl viologen 1,1′-dimethyl-1H,1′H-[4,4′] bipyridinylidene acts as a **strong donor** for Au contacts. This effectively reduces W_f and consequently the electron injection barrier.[129]

It is worth noting that, compared to the adoption of SAMs, the tuning of W_f through chemisorbed molecules has the clear advantage of not introducing a strong tunneling barrier thanks to hybridization of the energetic levels of the molecules and of the metal.[130]

It was recently shown that R_C can be strongly reduced for perylene-diimide derivatives based BC OFETs by inserting a CIL based on poly(3,4-ethylenedioxythiophene) poly(styrenesulfonate) (PEDOT:PSS) on top of Au contacts,[131] once again demonstrating the importance of semiconductor morphology. Despite an increase in the energy barrier, the improved crystallinity of the semiconductor on the contact area facilitates electron injection. A similar effect improves hole injection in pentacene.[132] These examples demonstrate once more how complex it can be to be predictive because of the interplay between many effects at the interface, where both energetic and morphological factors can contribute.

The literature proposes other examples of electron CILs, e.g., carbon-based CILs, where carbon nanotubes (CNTs) are adopted,[133] and sulphur-based CILs.[134]

So far, we have reviewed some examples of how CILs can be effective in tuning the injection for both holes and electrons in OFETs. Given the recent fast development of high mobility, solution-processed ambipolar semiconductors, capable of transporting holes and electrons equally well,[9,79–81,135] CILs could play an important role for the development of logic circuits based on these materials as they offer the possibility to engineer charge injection. Adoption of CILs would have the great advantage of allowing the fabrication of complementary logic based on a single ambipolar semiconductor. With this approach, the source and drain electrodes of all OFETs in the circuit could be realized with a single metallization layer, which is locally modified with a CIL to enhance the injection of one kind of carrier, while blocking the other, thus allowing easy definition of *n*-channel and *p*-channel areas. This approach is usually referred as “unipolarization” of an otherwise ambipolar OFET.[136] The semiconductor does not require fine patterning and can therefore be deposited with a simple coating technique. This was recently demonstrated for various solution processible, high-mobility ambipolar donor-acceptor polymers, such as poly(thienlylenevinylene-co-phthalimide)s (PTVPhI) derivatives and Polyera ActivInk P2100.[83] Good injection for holes was offered by bare Au contacts, which show very bad electron injection, while the formation of a thin CsF or Cs_2CO_3 CIL can suppress hole injection and strongly reduce R_C for electrons (**Figure 12**). The effective energy barrier for electrons is decreased thanks to the formation of Au-O-Cs or Au-F-Cs complexes[137] (a schematic representation of energy level alignment is given in Figure 12c and 12d). In the case of PTVPhI functionalized at the imide nitrogen with 2-ethylhexyl (PTVPhI-Eh) the shift is as high as 0.5 eV. This technique is compatible with simple spraying techniques, in which case *n*-type areas can be easily defined with the use of a shadow mask. It allowed simple fabrication of solution-processed complementary organic ICs, such as inverters with high-noise margins and ring-oscillators fabricated on flexible substrates. In the former case, a five-stage circuit oscillating at ~ 10 kHz was demonstrated, a result which would not have been possible without proper contact engineering.

2.4.4. Polymer and Other Carbon Based Electrodes

A strong trend is emerging in the literature that organic/organic interfaces might have some advantages for charge injection with respect to metal/organic interfaces.[138] π-π interactions in this case are quite relevant and can enhance interfacial morphology. In the previous section we have already described some of these interfaces, where organic CILs were adopted. Here we will review electrodes for OFETs completely based on polymer or other carbon based materials. These electrodes are an interesting alternative to metallic ones and can allow the realization of fully-organic devices. PEDOT:PSS, characterized by a relatively high W_f in the range of ~4.7 –5.2 eV, is the most known and adopted hole injector.[139,140] Given the much more limited extension of electron tail at the surface compared to a metal such as gold, the push-back effect is not severe in this case.[18] This allows for greater control of the energy level alignment at the interface. It can also constitute a good chemical interface for solution-processed organic semiconductors, as in the case of solution processible pentacene derivatives OFETs, where a good morphology and uniformity was achieved in the semiconductor at the interface with the electrode.[138] Despite steady progress in the conductivity,[141] PEDOT:PSS shows a higher sheet resistance than metallic electrodes, thus limiting its adoption as a self-standing electrode.[142] Of course there is still a lot of interest in developing high-conductivity polymer electrodes since they would offer the technological advantage of printability.[15,143] It was recently shown that with an enhanced conductivity of a few hundreds of Scm^{-1} it is possible to achieve $R_C \approx 30\ k\Omega \cdot cm$ in pentacene devices, comparable to R_C values obtainable with Au electrodes whilst still limited by the polymer conductor higher resistivity.[144] The further development of higher conducting formulations, bridging the gap towards metallic conductivity, would allow the development of fully scalable, high performance printed OFETs.

Another approach adopted for the reduction of PEDOT:PSS sheet resistivity, that for a Clevios PH500 formulation ranges from 200 kΩ/□ to 1 MΩ/□, is the blending with multi-walled CNTs (MWCNTs).[139] Sheet resistivity can be reduced to a few hundred Ω/□, allowing its successful adoption in the fabrication of complementary inverters.

PEDOT can also be made into a good electron injector, with low work function (4.3 eV), by doping with tosylate (PEDOT:Tos).[140]

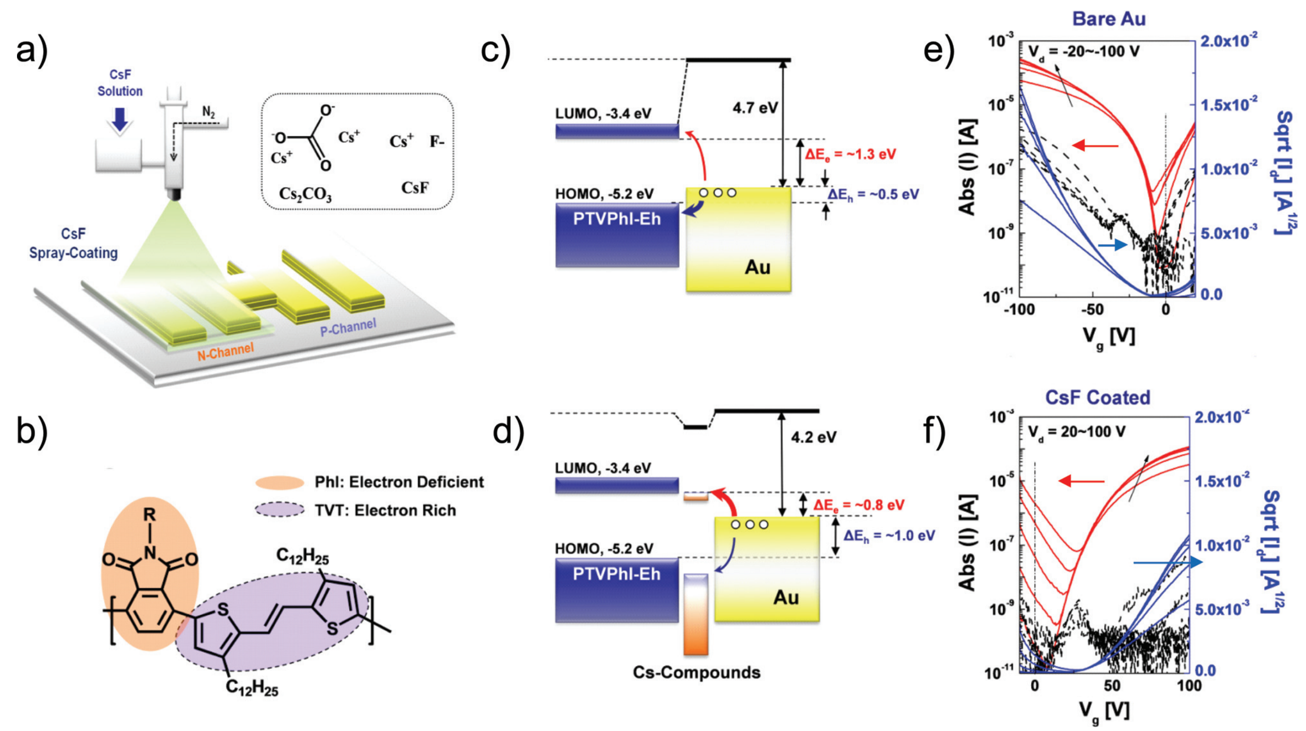

Figure 12. Contacts engineering for the unipolarization of OFETs based on high mobility ambipolar polymers. a) Spray coating can be adopted to locally modify Au electrodes with a Cs salt CIL. b) Chemical structure of a PTVPhI derivative. c) Au contacts are good holes injecting electrodes, so that the TGBC FET based on PTVPhI behaves like a *p*-channel device (e). d) If the Au electrodes are covered with a Cs salt CIL, the barrier for holes injection rises suppressing *p*-conductivity, while the barrier for electrons decreases, so that the FET behaves like a good *n*-channel device. Reproduced with permission.[83] Copyright 2011, American Chemical Society.

Given their high conductivity, carbon nanotubes are very good candidates themselves for solution-processible and flexible organic devices. Despite solubility problems, OFETs were demonstrated, both for the case of single-walled CNTs (SWCNTs)[145] and the case of multi-walled CNTs.[146] A recent approach allowed air-brushing of SWCNT, producing films with a sheet resistivity as low as 1 kΩ/□. These were demonstrated to be compatible with OFETs employing solution processed semiconductors such as P3HT.[147] R_C was comparable to or lower than in the case of Au electrodes depending on the semiconductors adopted and was similar to the case of chemical vapor deposited CNTs. In order to circumvent the solubility problems, PSS and other electrolytes have been adopted to make soluble formulations of MWCNT. Electrodes were fabricated based on PSS-wrapped MWCNST, showing the interesting feature of being photopatternable.[148] Their compatibility with solution processed TIPS-pentacene (TIPS-p) was demonstrated yielding high-mobility OFETs.

Graphene is very well known and it has been studied for its many interesting physical properties and applications.[149] It is also a very promising material for adoption as a transparent electrode in organic electronics, and in particular for the source and drain fabrication in OFET. A very good progress report has been recently published by S. Pang et al. on this topic which we refer the interested reader to.[150] Here we just wish to point out that graphene has been already adopted successfully for contacts in OFETs both as a CIL on top of metal electrodes (see Sect. 2.4.3) and as a self-standing electrode, showing a resistivity of only 200 Ω/□. The latter is compatible with solution processed P3HT devices, where ohmic injection has been suggested.[151] Monolayer graphene electrodes were adopted as well, though yielding lower performance solution-processed OFETs.[152,153]

2.5. Impact of Contact Processing Conditions

2.5.1. Vacuum Deposited Electrodes

As we described in Section 2.3.1, contact resistance effects are strongly dependent on the device architecture, and geometrical factors play an important role. This can be modeled and generalized, thus providing clear indications about the ideal configuration for optimizing charge injection. In this light BC and TC OFETs can be seen to be intrinsically different and usually yield quite different R_C values with the same semiconductor and contact electrode material. This happens because in the vast majority of the cases TC devices are realized in a staggered configuration, allowing current crowding effects to take place, while for BC both staggered and coplanar architectures are widely adopted. If technological constraints are taken into account however, the situation becomes complicated by the fact that the processing steps involved in the transistor fabrication strongly affect the final device performance. For example, aside from the geometrical difference between top and bottom contacts devices, one needs to take into account that evaporating a metal on top of an organic semiconductor creates a metal-organic interface physically different from the one formed by coating a metal electrode with a semiconductor. The nature of the interaction and its strength is expected to vary.[17] If we refer to the BC case, many details have to be accounted for. Firstly, low W_f adhesion layers (Cr and Ti most notably) are often employed in conjunction with noble metal contacts for hole-only bottom contact transistors. Since a low W_f metal is expected to establish a hole-depleted contact region, the thickness of the adhesion layers should be kept to a minimum, and it has been verified experimentally that values above 3 nm are detrimental.[154] Secondly, while defining source and drain, metals are subject to various processing steps (patterning, cleaning etc.) which can contribute to create spurious oxide layers, thereby degrading injection performance.[40] Thirdly, interactions between electrodes and dielectrics cannot be excluded in defining the contact energetics.[155]

Moreover, the final morphology of the coated semiconductor layer depends a lot on the surface energy it interacts with. The presence of contacts on the bottom and the possible contrast in surface energy between the contacts and the substrate has to be taken into account to achieve a uniform coverage with an ordered semiconductor, especially at the discontinuity edge between electrodes and substrate. Recently, in the case of BGBC devices the relevance of the physical height of the contact on top of the gate dielectric producing a sharp step was also underlined. It was demonstrated that by fabricating electrodes completely recessed in the gate dielectric it is possible to enhance the nanomorphology of a solution processed semiconductor not only at the edge of the electrode but over a larger area of the film, thus benefiting not only charge injection but transport as well.[156] On the other hand, in a staggered TC case, the semiconductor would be formed on a uniform substrate. There are also semiconductors which are known to show a poorly ordered morphology in the first few layers in contact with gold, while evolving towards higher order in the bulk and top surface. Chemisorption-driven defects at the interface with the metal are the basis of poor mobility and high R_C (>1 G$\Omega\cdot$cm) in *n*-type semiconducting α,ω-diperfluorohexylquaterthiophene (DFH-4T) based BC FETs, compared to the TC case (4–12 M$\Omega\cdot$cm).[157] Another factor to take into account is that air exposed bottom contacts (e.g., Au) are known to be subject of contamination by saturated hydrocarbons,[17] altering the nature of the interface.

In the case of top contacts the picture is completely different of course, as these are usually deposited by thermal evaporation or e-beam in vacuum. Here the level of vacuum in the evaporation chamber, especially in the case of reactive, low W_f metals, is important. It is very common practice to adopt vacuum levels of 10^{-6} mbar: although usually overlooked, these interfaces are completely different from those in UHV (< 10^{-8} mbar)[158,159] and even metals which are not strongly reactive such as Al are known to slightly oxidize in the surface region. For noble metals, reactivity is of course much lower or absent and therefore diffusion in organic semiconductors is higher. This leads to the formation of a contact which is inherently different from that in the previous case, as diffused metal clusters[160,161] can induce intra-gap states which alter the injection process. These defect states may favor injection by providing a hopping path through the energetic barrier at the interface[56,57] or through the bulk access region to the accumulated channel.[57] It is evident that in the case of TC a fair comparison between R_C in OFETs with different contact metals, or with contacts deposited with a different technique, cannot be exempted from a precise structural and chemical analysis to unveil the real nature of the interfaces.

To this extent, it was clearly reported in literature that the evaporation rate at which gold is thermally grown on top of pentacene affects the interface, which shifts from a neat, sharp interface yielding low R_C at high evaporation rates (7 Å/s), to a less abrupt, intermixed interface at low rates (0.2 Å/s) which is found to be detrimental for injection.[162] Higher rates were found to limit Au lateral diffusion in pentacene also in the case of e-beam evaporation.[163] This also demonstrates that the simplistic view where a more intimate contact achieved by fabrication of top electrodes is able to provide a lower R_C, cannot be generalized, as in more ordered and crystalline materials the local order may be disrupted, as an effect of excessive thermal budget for example. This was reported to occur for solution processed triethylsilylethynyl anthradithiophene (TES-ADT), which is susceptible to thermal damage, due to the formation of weakly stable crystals: local melting at top-gold/semiconductor interface was found to drastically increase R_C (10.2 M$\Omega\cdot$cm).[164] Lower R_C (590 k$\Omega\cdot$cm) can be achieved in the same device if soft-contact patterning methods are adopted to form contact electrodes. The adoption of protective CILs was also found to relax this effect.[165]

2.5.2. Solution Processed Electrodes

It is desirable to shift from vacuum deposited to solution processed electrodes, therefore enabling the printability of each component of an OFET. We have already seen in Section 2.4.4 that there are some polymeric and other carbon based conductors which allow processing from solution. The problem is developing a suitable ink formulation for the required printing technique which can yield a conductive enough contact. As the landscape of printed electronics is very broad, here we will be limited to the case of inkjet printing.[166,167] This poses a limitation in perspective but we believe that examples in this sector show some general features that are interesting for all the field of solution-processed OFETs.

We showed how PEDOT:PSS is in principle highly compatible with OFET fabrication, despite the fact that its limited conductivity can negatively affect especially short-channel devices. To avoid this, more conductive formulations, obtained through blending with CNTs or metal nanoparticles[168] are being developed, while high conducting metallic inks have also been developed as an obvious alternative. The aim is to lower the costs (e.g., shifting from Au based inks to Ag and Cu based ones[169–171]) and the sintering temperature, so that they can be compatible with polymeric substrates. As a result of cost and technological constraints, the range of materials tends to narrow. For example highly reactive metals such as Ca are not good candidates for an ink, due to the intrinsic problems in developing a stable formulation. Although Cu is much less reactive, its tendency to form an oxide has made it more complicated to develop inks based on this comparatively cheap element. Therefore the simple approach of energy level matching between the contact and the semiconductor to optimize injection, which is inherently very limited as we showed, becomes even less applicable. Moreover, given the specific nature of the metallic inks, which require a sintering process, different effects at the surface can occur due to: *i*) the original ink formulation (e.g., a nanoparticle based ink or a metallic complex[166]); *ii*) the sintering time and temperature; *iii*) the environment where sintering is performed (e.g., in air or controlled atmosphere). As a result carbon based residuals or contaminants might be present on the contact surface and this has of course an effect on its W_f.[172] It was also reported how the W_f of a printed Ag contact sintered in air can be wildly different from the W_f of a thermally evaporated Ag electrode.[173]

Coming to applications, the adoption of a noble metal such as Au for the development of the first metallic inks favored the fabrication of hole injecting contacts, which also turned out to be compatible with sub-micrometric printed OFETs devices (**Figure 13**a).[87,174] Integration of less expensive Ag printed contacts in a *p*-type, BC OFET is possible even without a surface treatment,[175] though this is needed for optimized OFET performance. This is the case in TIPS-p devices where the surface energy contrast of the electrode with respect to the substrate is removed by a F_4TCNQ blanket treatment favoring an improved film morphology (Figure 13b).[172] Ag printed lines showed compatibility with *n*-type BC OFETs,[168] as well as improved performance with respect to Au contacts in sub-micrometric printed devices.[176]

Printed TC OFETs are much less common in the literature due to the possible damage of the top semiconductor surface caused by the solvents present in the ink. A way to solve this would be to develop formulations where the solvents and additives are completely orthogonal to the semiconductor. In any case, alternative processing conditions allow the realization of high-performance TC OFETs, as in the case where very small Ag droplets, favoring fast drying of the solvent, are printed on top of a pentacene layer. It was demonstrated that for a droplet volume below 2 pl the reduction of the OFET performance due to the solvent can be drastically reduced thanks to a smaller R_C with respect to the case of higher droplets volume.[177] The same concept allowed the demonstration of downscaled devices thanks to a sub-femtoliter inkjet printing system which allowed a ~ 1 μm long Ag channel to be printed on top of both *p*- and *n*-type semiconductors (Figure 13d).[14]

Of course if it were possible to develop carbon based inks (e.g., CNTs, graphitic materials) compatible with low cost printing processes, they would represent an ideal alternative

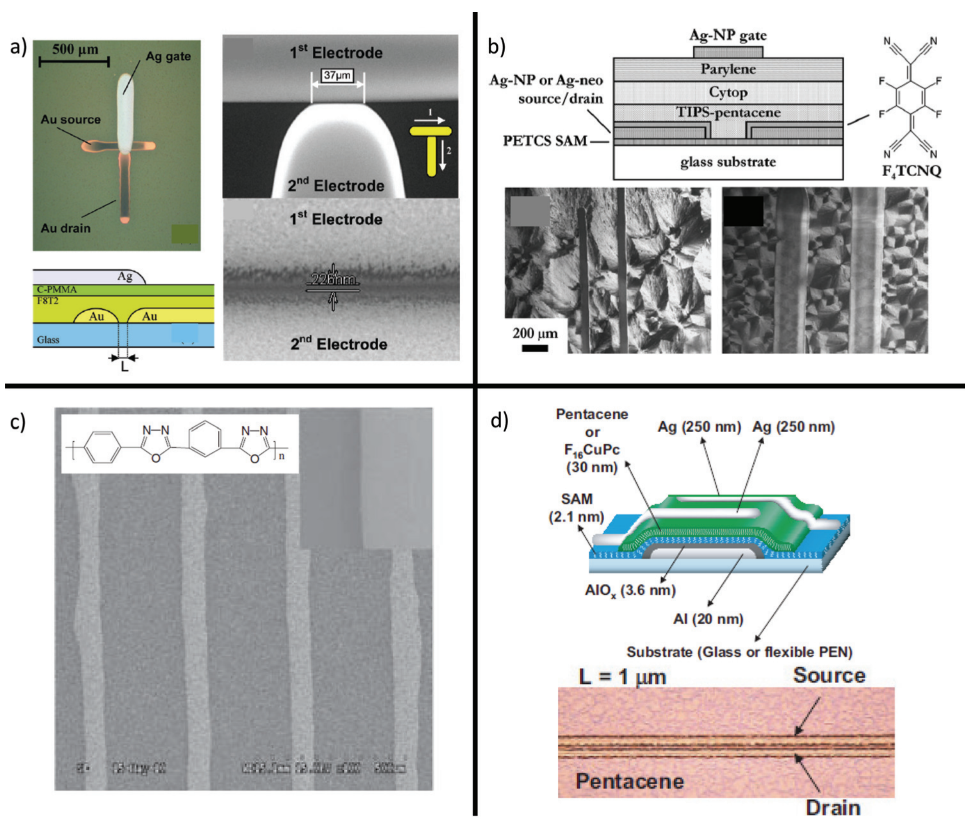

Figure 13. a) Mask-less, solution processed OFET with sub-micrometric inkjet printed contacts obtained thanks to the self-aligned printing (SAP) technique. Good holes injection in F8T2 is achieved by PFDT treatment of the Au electrodes. Reproduced with permission.[174] Copyright 2010, American Chemical Society. b) Solution processed OFET based on TIPS-p where injection from inkjet printed Ag electrodes is enhaced by a blanket treatment with F_4TCNQ. Reproduced with permission.[172] Copyright 2009, American Institue of Physics. c) SEM images of the inkjet printed graphite-like electrodes obtained by pyrolized PPOD (inset), which provides good injection both in *n*- and *p*-channel OFETs. Reproduced with permission.[178] Copyright 2011, WILEY-VCH Verlag. d) One of the very few examples of inkjet printed TC devices. Here the resolution is enchanced by adoption of a subfemtoliter inkjet printer. Reproduced with permission.[14] Copyright 2008, National Academy of Sciences.

to metallic inks, given their high conductivity and enhanced compatibility with organic semiconductors. Recent progress has been achieved in this direction with the demonstration of inkjet printed graphite-like electrodes obtained from pyrolyzed poly(l,3,4-oxadiazole) (PPOD) (Figure 13c). Despite a strong limitation due to the very high processing temperature needed for pyrolization (~800 °C), BC OFETs with limited R_C and improved mobility with respect to Au were obtained both for *p*-type and *n*-type transistors as a cumulative result of morphological and energetic effects, attesting to the better compatibility of such electrodes with organic semiconductors.[178]

3. Contact Resistance Extraction Methods

3.1. Methods Based on Current/Voltage Measurements

Injection limitations can be assessed by means of *static* or *dynamic* electrical measurements. In the former case one exploits the fact that non-ohmic contacts alter the dependence of the current on: *i)* the channel length *L* (Section 3.1.1.); *ii)* the applied gate to source and/or drain to source voltage (Section 3.1.2.). Advantages of combining both dependencies are introduced in Section 3.1.3. Fitting and iterative approaches are reported in Section 3.1.4. Dynamic methods are briefly reviewed in Section 3.1.5.

3.1.1. Exploiting the Scaling of Current with Channel Length

The Transfer (or Transmission) Line (or Length) method (TLM) has acquired large popularity in the literature thanks to the ease of application and limited effort in interpreting the results.[40] At the core of the TLM lies the observation that contacts are interface-related issues and as such are expected to be independent on *L*, whereas the channel resistance is obviously a function of *L*. Thus, the scaling of current in a set of FETs with different *L* is exploited to extract FETs parameters.

In the linear regime, an OFET device can be seen as the series connection of an ideal FET, characterized by its *L*-dependent channel resistance $R_{ch} = L/[\mu C_{ox} W(V_{GS}-V_T)]$, and of the source and drain contacts, modeled as an *L*-independent contact resistance R_C. Hence, for a fixed V_{GS}, if the total resistance R_{TOT}, corresponding to the experimentally measured ratio $V_{DS}/I = R_{ch}(L) + R_C$, is plotted against *L*, a straight line is obtained, whose slope is proportional to $[\mu(V_{GS}-V_T)]^{-1}$ and whose extrapolated *y*-axis intercept is R_C. In the framework of the so-called *gated*-TLM (gTLM), this procedure can be repeated by varying V_{GS}, and the reciprocal slopes can be plotted against V_{GS}, thus obtaining (under the simplifying approximation of constant mobility) a straight line whose slope and extrapolated *x*-axis intercept yield μ and V_T respectively (**Figure 14**).

The fundamental drawback of gTLM is that it critically relies on the capability of producing a set of devices with different *L* characterized by the same parameters (most notably μ and V_T): process and sample-to-sample variations result in data scattering which increases the uncertainty in the fit and deteriorates the accuracy of the method.[179] A partial workaround has been suggested based on the observation that μ and V_T tend

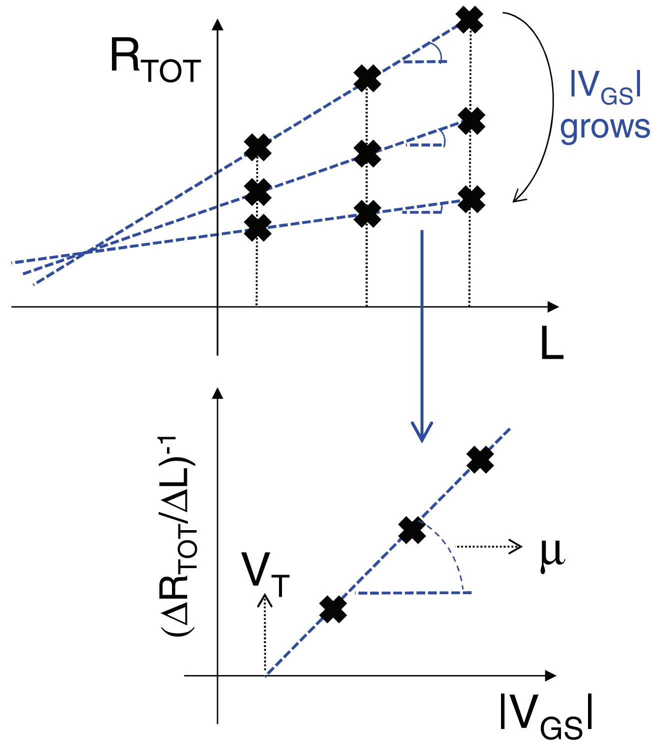

Figure 14. Qualitative sketch of the application of the TLM method (on ideal non scattered data). The total measured resistance is plotted against channel length *L* for various gate voltages (top). In the case of contact resistance $R_C = R_{C0} + \alpha/[W(V_{GS}-V_T)\gamma+1]$, a bundle of straight lines is obtained. The center of the bundle lies at $(-\mu_0 C_{ox}\alpha, R_{C0})$. The reciprocal of the lines slopes is equal to $\mu_0 C_{OX} W(V_{GS}-V_T)\gamma+1$, hence by plotting it versus $V_{GS,}$ the mobility and the threshold voltage can be extracted (on the bottom, the case of a constant mobility, viz $\gamma = 0$).

to be more dispersed than R_C. By plotting R_{TOT}/L versus $1/L$, a straight line is obtained whose slope is proportional to R_C: doing so, R_C is estimated more accurately than in the case of classic TLM.[180]

The gTLM can also be used when R_C is dependent on V_{GS}. Generally speaking the *y*-axis intercept becomes V_{GS}-dependent and can be plotted versus V_{GS} to obtain $R_C(V_{GS})$. A commonly adopted model for R_C consists of a sum of a constant term R_{C0} and of a V_{GS}-dependent term: $R_C = R_{C0} + \alpha/[W(V_{GS} - V_T)]$ (the parameter α having uncertain physical meaning).[74] In this case it can be shown that by varying V_{GS}, the plots of the total resistance versus *L* constitute a *bundle* of straight lines, whose center is generally located in the second quadrant at coordinates $(-\mu C_{ox}\alpha, R_{C0})$. When the center lies on the negative (positive) part of the *x*(*y*)-axis, the constant (V_{GS}-dependent) term is negligible.[179] The same holds true when a V_{GS}-dependent mobility in the form of a power law is considered, $\mu = \mu_0(V_{GS} - V_T)\gamma$, and when $R_C = R_{C0} + \alpha/[W(V_{GS} - V_T)\gamma+1]$. This dependence of mobility on gate voltage is predicted to occur both in the multiple trapping and release model[181] and in the hopping transport in an exponential tail model.[179,182]

Recently, the gTLM has been reformulated to take into account current crowding in staggered devices (with finite electrode width and a purely resistive transport along *y*-direction).[74]

If contacts cannot be modeled as resistances, which means that their current/voltage relation is non-linear, the plot of R_{TOT} versus L is no longer a straight line but acquires a convexity thus invalidating the application of the gTLM (as occurs for instance in the current crowding model with SCLC along y-direction[68]). The fault originates from biasing all the FETs in the selected set with the same applied voltage: since V_{DS} is the sum of the voltage drop on the channel and on the contact, viz. $V_{DS} = R_{ch}(L)I(L) + V_C[I(L)]$, it turns out to depend non-linearly on L through V_C. Hence upon dividing V_{DS} by $I(L)$ to obtain $R_{TOT}(L)$, the second addendum remains L-dependent and the plot of R_{TOT} versus L does not yield the expected straight line. To deal with this situation, it would be more convenient to bias the FETs set with the same current I (and the same V_{GS}), allowing V_{DS} to vary, viz. $V_{DS}(L) = R_{ch}(L)I + V_C(I)$: in doing so V_C would be the same for all devices, and in a V_{DS} versus L plot, the y-axis extrapolated interception would yield the V_C pertaining to the biasing current. By repeating the procedure with various biasing currents, a plot of V_C versus I could be constructed. It should be noted that this technique, and indeed all the techniques based on current-voltage measurements, are only sensitive to V_C, the sum of the voltage drop on source and drain. In case of non-linearity, the analysis is meaningful if V_C can be unambiguously traced back to only one of the contacts by means of independent measurements, or if the injection physics is well known; otherwise no relevant conclusions can be drawn.

Another approach useful in non-linear injection is outlined in Ref.[58]. The transistor current is modeled as $I(L-d)/(WC\mu) = (V_{GS}-V_T)(V_{DS}-V_C) -0.5(V_{DS}{}^2 - V_C{}^2)$, under the assumption that V_C drops entirely at the source side (d is the contact phyisical extension and is assumed to be negligible with respect to L). It should be noted that the full equation for FET operation is used, and hence this method is not limited to the linear regime where it is usual to neglect second order terms (see for instance the previously outlined gTLM). The current equation can be solved for V_C to obtain $V_C(I)$, but this requires knowledge of μ and V_T. To this extent, a set of FETs with different channel lengths is measured, and μ and V_T are chosen so that $V_C(I)$ curves from FETs with various L collapse into a single curve, exploiting the fact that $V_C(I)$ is an interfacial property and as such is the same for all devices. The main advantage of this approach is that no *a priori* assumption is made on the functional form of $V_C(I)$, but on the other hand the task of finding the right μ and V_T might be time consuming (in addition to the aforementioned difficulty in producing a set of identical FETs with different L). In the literature, it has been suggested that V_T and μ extracted from a saturation regime should be used, based on the hypothesis that contact resistances are negligible in such a regime.[54]

3.1.2. Methods Exploiting the Functional Dependence of Current on V_{GS} and V_{DS}

It is always possible, starting from a certain model with a certain number of unknown parameters, to fit experimental data by proper choice of parameters obtained through an iterative optimization. Obviously, the larger the number of unknowns, the harder it is to perform the fitting. In addition, the chosen model might be inadequate for fitting the whole experimentally explored variable space: to ascertain this and to avoid inaccuracies in extracted parameters, detailed refinements of the fitting region might be necessary. To mitigate this scenario, it is often advisable to adopt strategies that lower the degree of complexity of the problem. This can be done if, by applying proper mathematical operators to the measured quantities, auxiliary functions can be computed with a simplified functional dependence and with a reduced set of unknowns.

To illustrate this, we start with a simple example and consider the linear transfer characteristic of a FET characterized by constant contact resistance R_C. It can be easily shown that $I = \frac{K V_{DS}(V_{GS}-V_T)}{1+K R_C(V_G-V_T)}$, where $K = \mu C_{OX}W/L$. Now if one considers the auxiliary function $z = \frac{I^2}{I'}$, (where the prime indicates first derivative with respect to V_{GS}), it can be shown that $z = K(V_{GS} - V_T)^2 V_{DS}$: z is independent on R_C and is a simple quadratic function of V_{GS}. This method, also known as the Y-function method, was originally introduced for constant mobility[183,184] and for specific functional dependences of mobility,[185] and later was shown to be valid for an arbitrary dependence of the mobility on V_{GS} and termed the differential method (DM).[75,179] In the important example of mobility in the form of a power law, $\mu = \mu_0(V_{GS}-V_T)\gamma$, z becomes a power law of $V_{GS}-V_T$, viz. $z = \frac{K}{\gamma+1}(V_{GS} - V_T)^{\gamma+2} V_{DS}$, and further complexity reduction is possible by the so called H-function method.[179,186] In fact, the auxiliary function $w = \frac{\int_0^{V_{GS}} z(\widetilde{V_{GS}})\, d\widetilde{V_{GS}}}{z(V_{GS})}$ is a simple linear function of $V_{GS}-V_T$, viz. $w = \frac{1}{\gamma+3}(V_{GS} - V_T)$, with just two unknowns: γ and V_T (**Figure 15**). Once w is fitted as a straight line in a w versus V_{GS} plot, it is straightforward to go back and calculate the remaining unknowns.

The advantages of DM are: *i)* it is applied to a single device, so it avoids the problems related to FET parameter dispersion which can affect the accuracy of gTLM; *ii)* in case the model requisites (μ in the form of a power law of V_{GS} and constant R_C) are satisfied only in limited ranges of V_{GS}, it is easy, and it is easier than in a pure fitting approach, to isolate where these constraints are met: in the validity regions z versus V_{GS} has to behave as a straight line and its slope has to lie between 0 and 1/3 (provided that γ can range from infinity to 0). Hence data ranges where the method cannot be applied can be easily identified, and they do not compromise the parameter extraction elsewhere thanks to the fact that DM is applied pointwise, in contrast to fitting which minimizes the mean square error between experimental data and the fitting curve over a more or less arbitrarily chosen data range.

A variant of DM was also developed in the saturation regime, where the resistance on the source side diminishes the voltage drop across the actual channel, $V_{GS}' = V_{GS}-R_S I$, and the drain-side resistance has no role as long as the device remains in saturation regime despite the voltage drop on R_D. The auxiliary function to be used in this case is $z^{-1} I = \frac{I''}{I'^2} I = \left(\frac{\gamma+1}{\gamma+2}\right)(1 - R_S I')$, where $z = \frac{I'^2}{I''}$. Plotted against dI/dV_{GS}, $z^{-1}I$ is a straight line whose fitting yields R_S and γ.[179] A relevant feature of the DM is that it can be applied under saturation even though an explicit expression for current is missing in this regime. It has to be noted that both the first and the second derivatives of current with respect to V_{GS} are needed, thus requiring measurements with very low degree of noise (because of the high-pass filter effect of the derivative operator). The DM can also be used in

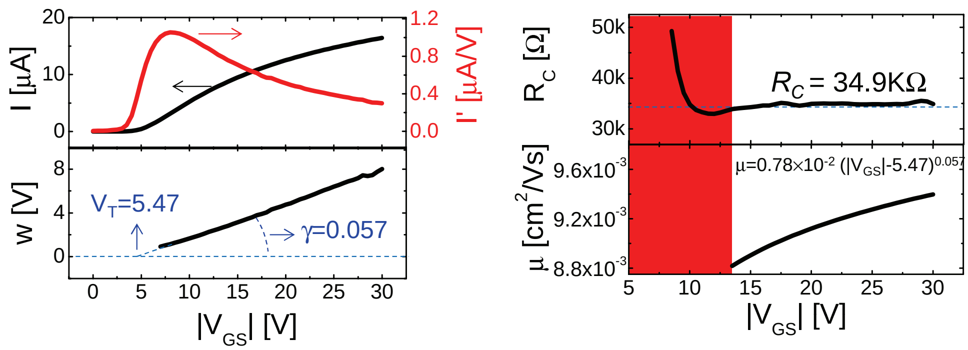

Figure 15. Example of application of the differential method in the linear regime to a BCBG P3HT-based transistor. On the top left: the downward concavity of current versus $|V_{GS}|$ curve suggests the presence of contact resistances. On the bottom left: the quantity w versus $|V_{GS}|$ behaves as a straight line. From its slope and x-axis interception γ and V_T are extracted. On the right: extracted contact resistance and mobility. The analysis is valid only in the region where R_C is constant with V_{GS}: in the red area this condition is not met and the method cannot be applied. Data reproduced with permission from Ref. [179]. Copyright 2007, American Institute of Physics.

the case of mild non-linear injection, provided that for a fixed V_{DS} the contacts can be reasonably approximated as resistors, or in other words that their current voltage characteristic can be linearized around each V_{DS} bias point. By applying the DM for different V_{DS}, it is possible to reconstruct a dependence of R_C on V_C.

While the DM uses transfer characteristics either in the linear or in saturation regime, the **transition voltage method** (TVM)[187] exploits the complete output characteristic curve. It is based on the following experimental observations of the authors: *i)* R_C is a decreasing function of V_{GS}; *ii)* due to *i)*, the voltage drop on contact resistance is almost independent of V_{GS}. It has to be underlined that before the application of TVM, hypotheses *i)* and *ii)* have to be verified by other independent methods. If TVM can be applied, first V_T is extracted from the linear regime by extrapolating the x-axis intercept of the transfer characteristic curve: this requires the V_{GS} dependence of the mobility to be negligible so that the transfer characteristic curve can be reliably approximated as a straight line. Secondly, the output characteristic curve in the linear regime is extrapolated until it intercepts the current value in the saturation regime: this occurs on the x-axis at the so called transition voltage V_{tr} (**Figure 16**). Under the additional hypothesis - indeed quite stringent and not so often verified in OFETs - that contact resistances in linear and saturation regimes are the same, the ratio between current in the linear regime extrapolated at V_{tr} and current in saturation regime is computed. This auxiliary function does not depend on the mobility, and can be solved for the only unknown left which is R_C, viz. $R_C = (V_{GS}–V_T)^{0.5}[(2V_{tr})^{0.5} – (V_{GS}–V_T)^{0.5}]/I_{sat}$. By repeating the procedure for various V_{GS}, the V_{GS} dependence of R_C can be obtained.

To handle and to check whether **strong non-linearities** are present, it is possible to take advantage of the operator $\left[\frac{\delta F(x)}{\delta x} - \frac{F(x)}{x}\right]$, which removes linear terms.[188] The experimentally measured quantity of interest here is the inverse of the output characteristic $V_{DS}(I)$ in the linear region. By inspection of the power series expansion of $V_{DS}(I)$, viz. $V_{DS}(I) = I(R_{ch} + R_C) + \sum_{i=2}^{n} I^i A_i$, an equivalent circuit can be constructed where the various *addenda* are connected in series. The FET contributes only to the linear term via R_{ch}, while the non-linear contact contributes to the linear term via R_C, and also to

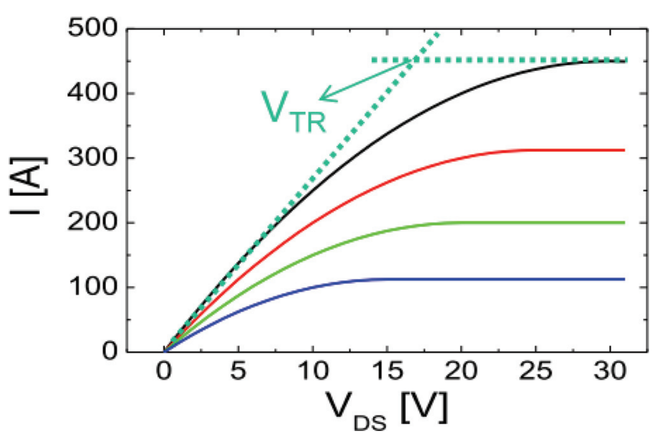

Figure 16. Sketch of extraction of the transition voltage V_{TR} from output characteristic curves. Redrawn after Ref. [187].

higher order terms A_i. The auxiliary function $\left[\frac{\delta V_{DS}}{\delta I} - \frac{V_{DS}}{I}\right]$ contains only terms originating from the non-linearity of the contacts, viz. $\left[\frac{\delta V_{DS}}{\delta I} - \frac{V_{DS}}{I}\right] = \sum_{i=2}^{n}(i-1)A_i I^{i-1}$. By fitting $\left[\frac{\delta V_{DS}}{\delta I} - \frac{V_{DS}}{I}\right]$, the coefficients A_i, which are responsible for the non-linear injection, can be determined. Then by subtraction the voltage drop on the linear terms R_{ch} and R_C can be calculated. Finally, in order to discriminate between R_{ch} and R_C, the DM can be applied (repeating the procedure with operator $\left[\frac{\delta V_{DS}}{\delta I} - \frac{V_{DS}}{I}\right]$ for various V_{GS}), or the TLM can be adopted (repeating the procedure for various channel lengths).

3.1.3. Mixed Methods

To yield more information and/or to reduce the number of model assumptions, the dependence of current on both V_{GS} and L can be exploited. Combining **DM** and **gTLM** is very effective in case of contact resistances given by the sum of a constant term and of a V_{GS}-dependent term in the form $R_C = R_{C0} + \alpha/[W(V_{GS} - V_T)]$.[179] This case could be also treated by using only gTLM, but this method is vulnerable to parameter dispersion of FETs. To overcome this, firstly R_{C0}, V_T and an *apparent mobility* μ_{app} are extracted *device by device* by means of the DM. The extracted mobility is *apparent* because is still affected by the V_{GS}-dependent term of the contact resistance according to $\mu_{app} = \mu/ [1+ (\mu\alpha C_{ox}/ L)]$. Secondly, a TLM approach on the apparent mobility is applied to extract the *intrinsic mobility* and the V_{GS}-dependent term of the contact resistance (by plotting μ_{app}^{-1} versus L^{-1} a straight line is obtained yielding μ^{-1} from the extrapolated *y*-axis intercept and α from the slope). This approach is advantageous with respect to applying only gTLM, because R_{C0} and V_T are extracted from each single device by means of DM and are thus allowed to be dispersed. An analogous approach can be adopted when the mobility is a power law of V_{GS}, viz. $\mu = \mu_0(V_{GS} - V_T)\gamma$, and R_C has the form $R_C = R_{C0} + \alpha/ [W(V_{GS} - V_T)\gamma+1]$: in this case the device by device application of DM yields R_{C0}, V_T and γ.

It must be noted that in these cases DM is applied to a device affected by the sum of constant and V_{GS}-dependent contact resistances, which might seem to contradict the previous statement that DM can be applied only when *constant* contact resistances are present. Indeed we exploit the fact that a *purely* V_{GS}-dependent contact resistance in the form of a power law ($R_C = \alpha/ [W(V_{GS}–V_T)]$), results in an FET which is *apparently unaffected* by contact resistance as far as the functional dependence of I on V_{GS} is concerned, viz. $I \propto (V_{GS}–V_T)V_{DS}$. This occurs because R_{ch} and R_C depend on V_{GS} in such a way that V_{DS}' turns out to be a *constant* fraction of V_{DS}, viz. $I \propto (V_{GS} - V_T)V_{DS}' \propto (V_{GS} - V_T)V_{DS}$ (whereas in the general case V_{DS}' is a V_{GS}-dependent fraction of V_{DS}, and this alters the functional dependence of current on V_{GS}). The result is that $R_C = \alpha/ [W(V_{GS}–V_T)]$ only affects a multiplicative factor in the current expression. The same holds true when $\mu = \mu_0(V_{GS}–V_T)\gamma$ if the *purely* V_{GS}-dependent contact resistance is in the form $R_C = \alpha/ [W(V_{GS}–V_T)\gamma^{+1}]$.

From the considerations above, a number of *caveats* can be given: *i)* a textbook like functional dependence of I on V_{GS} does not rule out the presence of contact resistances *per se*; *ii)* if by applying DM negligible contact resistances are extracted, this means that R_{C0} is negligible, but nothing can be said about V_{GS} dependent terms until a gTLM test is performed; *iii)* DM-extracted μ_0 showing a dependence on L should be looked upon suspiciously and tested with TLM as they might be apparent mobilities.

Other approaches which can be found in the literature follow. In Ref.[189] firstly gTLM is used to extract the mobility; secondly transfer characteristic curves are iteratively fitted. An arbitrary dependence on V_{GS} is allowed for the mobility, while a power law dependence on V_{GS} is assumed for contact resistances.

In ref.[190] the FET is modelled as an intrinsic ideal transistor in series with a so-called contact transistor. The parameters of the former are modelled on long channel devices, where contacts effects are expected to be negligible; the parameters of the latter are extracted from short-channel devices.

3.1.4. Iterative Approaches

Another class of approaches relies on a first rough estimation of parameters using simple models, followed by (sometimes iterative) refinement.

In Ref.[191] contact resistances are disregarded at low V_{GS} under the hypothesis they are negligible with respect to R_{ch}. Hence the H-function method is applied to extract mobility and threshold voltage. Then R_C is taken into account and extracted for large V_{GS}. Within this context, non-linear injection is modeled as a diode in series with the FET: firstly FET parameters are extracted where non linearity is negligible (at relatively high applied V_{DS}, when the diode dynamic resistance is low), and then the regions showing high non linearity are modeled (for a critical discussion on the use of diodes to model non-linear injection the reader is referred to Ref.[47]).

In Ref.[192] the equation for the linear region $I(L–d)/ (WC\mu) = (V_{GS}–V_T)(V_{DS}–V_C) – 0.5(V_{DS}^2 - V_C^2)$ is rewritten introducing the parameter $\alpha = V_C/ V_{DS}$. Fitting is performed iteratively using the mobility extracted from the saturation regime (under the hypothesis the contact resistance is negligible in saturation) and $\alpha = 0$ as starting guesses.

3.1.5. Dynamic Techniques

Finally we give a brief account of **dynamic techniques** for contact resistance extraction, which exploit impedance spectroscopy to model the device behaviour under sinusoidal modulation. In Ref.[71] impedance spectroscopy was applied to a MIS capacitor, modelled with lumped elements, in order to measure the transverse conduction across the semiconductor and thus to obtain an estimate for R_y in the framework of the current crowding model.

In Ref.[193] a FET with grounded source and drain was modelled in the framework of transmission line RC network: the fitting of the experimentally acquired impedance signal proved to be sensitive to contact phenomena, modelled as a contact resistance and contact capacitance in parallel (the latter attributed to a contact depletion region). The modelling of non-ohmic contacts in terms of parallel RC network is confirmed in Ref.[194] where the impedance could be modelled without the contact RC network upon making the interface ohmic by local doping.

3.2. Experimental Methods for Direct Extraction

In this Section we will review the most common and reliable experimental approaches adopted for the (almost) direct measurement of R_C. These methods are very relevant as they provide a cross-check for the extraction methods described in Section 3.1, allowing the quantification of R_C with no or a limited amount of assumptions and more importantly permitting the discrimination between R_S and R_D. A simplified classification of these methods involves on the one hand methods to extract point-wise information with local probes along the channel, as in Gated Four-Point Probe (gFPP), and on the other scanning techniques, where the whole length of the channel is investigated.

3.2.1. Gated Four-Point Probe

The gFPP technique was firstly developed for amorphous Si devices[195] and it derives its name from the very well known four-point probe technique adopted to measure sheet resistivity, applied here to a gated device such as a FET. Two small high-impedance probes are patterned in the channel of the device and used to measure the channel potential at two precise points

(**Figure 17**). In BGTC and BGBC devices the probes come directly into contact with the channel area where, through their own contact resistance, they are charged to the local channel potential. In the TGBC case the probes are in contact with the high-resistive bulk semiconductor and the charging of the probes to the correct channel potential occurs through the least resistive path connecting them to the channel.

In order not to affect the device behavior, the total probe area, which is equipotential, should be limited and the probe should protrude into the channel for a very limited length.[48] Another very important requirement is that the gated area, therefore the gate contact or the semiconductor, need to be carefully patterned in the channel area. This is because the probes charge to a potential which is an average of all the area where there is an overlap between the probes and the gated semiconductor, with the effect that, if either the semiconductor or the gate is not properly patterned, areas outside the channel would be probed.[48,67]

In the linear regime, the voltage drops at the source (ΔV_S) and drain (ΔV_D) contacts can be extracted by linear extrapolation under the assumption that the mobility is constant with charge concentration.[57] Therefore R_S and R_D can then be simply obtained from Ohm's law as $\Delta V_S/I_D$ and $\Delta V_D/I_D$ for source and drain respectively. This method is clearly not valid in the saturation regime, where pinch-off is reached at the drain side and the field profile is highly non linear. A more evolved method was developed to take into account the actual channel voltage profile and so extend gFPP to high lateral drain voltages and to cases where the mobility is gate voltage dependent.[67] The method is based on an algorithm to extract the gate-voltage dependence of the mobility from the measured data, and on adopting this to numerically solve the potential profile between the two probes and between the probes and each contact.

gFPP was in general applied to the case of pentacene[57,196] or other evaporated small molecules devices in the field of organic FETs,[197] and then successfully employed with various solution processed FETs.[67,75,89,198] In the previously described evolved methodology, gFPP is especially powerful as it allows an almost direct extraction of R_C in all the OFET configurations, although it requires the careful fabrication of probes integrated in the device and a patterned gated area. Alternative approaches rely on the adoption of an AFM conductive tip to acquire point-wise, local information of the potential in the channel avoiding the fabrication of integrated probes in the device.[199,200]

3.2.2. Scanning Techniques

Much more detailed information on the potential profile along the channel of an OFET can be obtained by scanning potentiometry techniques, among which SKPM is the most well known and adopted for the purpose of R_C extraction.[201,202] SKPM is a non-contact potentiometry technique where a conducting, vibrating probe is scanned along the channel surface of a transistor while minimizing the electrostatic force between

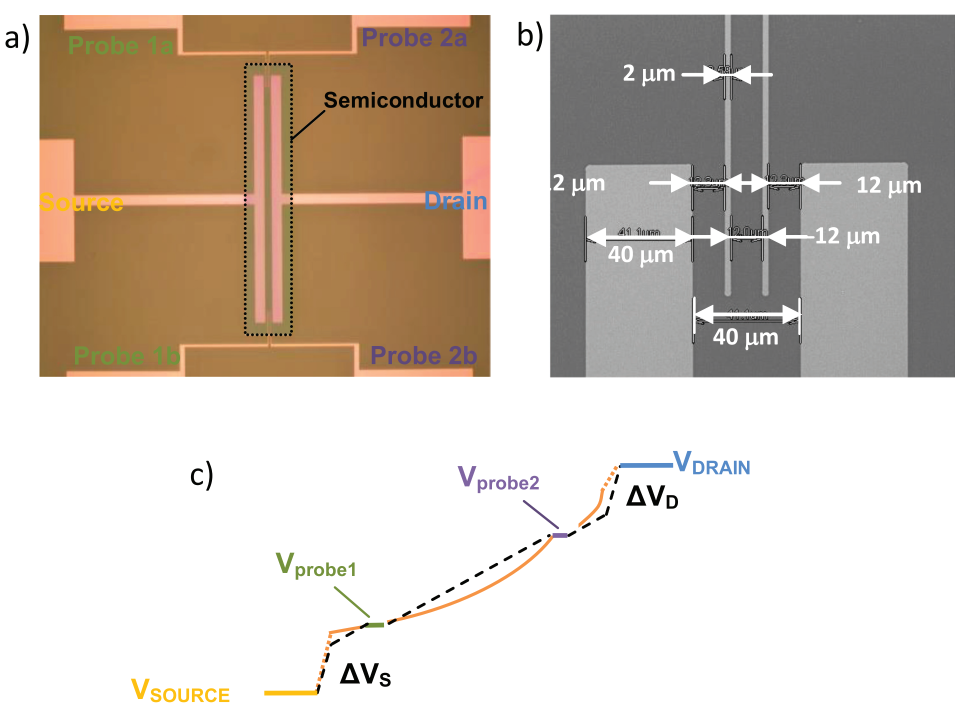

Figure 17. a) Micrograph of a bottom gold pattern to be adopted in gFPP measurements: on both sides of the channel a couple of probes are defined, 50 µm protruding in the channel area, around which the semiconductor is patterned. b) A SEM image shows the detail of a pair of probes, 2 µm wide, on the top side area of the channel. c) A simple scheme of the voltage drop along the channel: based on the probed V_{probe1} and V_{probe2} values, the dashed black line illustrate the simple linear extrapolation; by adopting an advanced model and by taking into account the actual charge concentration along the channel,[67] more complex voltage profiles can be accounted for (a schematic representation is indicated by the solid line).

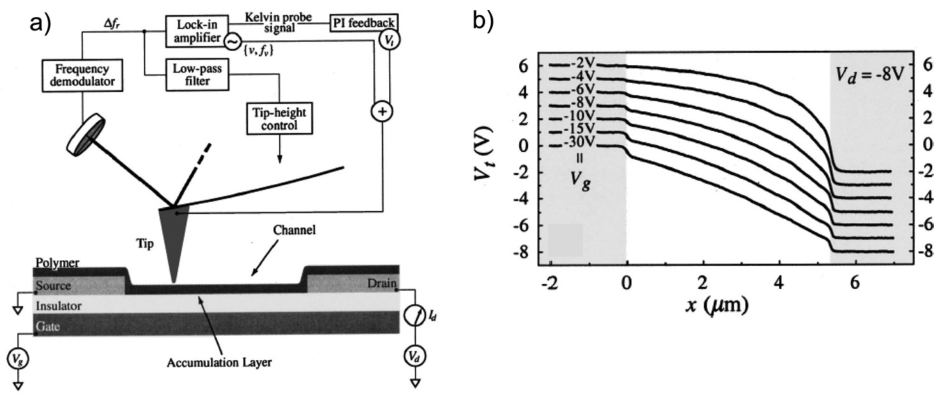

Figure 18. a) Schematic representation of the SKPM setup. b) Examples of potential profiles acquired on P3HT BGBC FETs at different gate voltages, where a 1 V offset is applied to consecutive traces for clear visualization. Reproduced with permission.[203] Copyright 2002, American Institute of Physics.

the tip and the surface (**Figure 18**). This is applicable both to a BGTC architecture and to a BGBC one, as it was demonstrated that the surface potential in a thin-film organic transistor actually corresponds to the potential along the buried channel at the semiconductor/dielectric interface, of course once a correction for the W_f of the tip is taken into account.[203] Here there is no need for the integration of probes as in gFPP and a full potential profile from source to drain can be obtained instead of just a couple of points. Therefore the potential drop at the contacts can be finely visualized, as the lateral resolution of this technique can be better than 50 nm.[204] SKPM is therefore a very powerful technique for characterizing contacts effects in FETs.[205] The drawback of SKPM is the need for highly controlled test conditions (e.g., ultrahigh vacuum) to completely remove artifacts such as those associated with water absorption on the surface, and the required accessibility of the top polymer semiconductor surface, thus precluding analysis of top-gated devices.

Other potentiometry techniques were also adopted, such as atomic force microscopy with a conductive tip.[199,200,206] This approach retains in principle the same resolution and detail of SKPM, but it is more delicate as the required contact between the tip and the semiconductor can lead to the damage of the semiconductor surface.

Of a slightly different nature is the adoption of a technique based on the Second-Harmonic Generation (SHG) induced by the microscopic electric field.[207] This is not strictly a scanning technique since the fundamental light is uniformly irradiated on the sample and the SHG signal, from which the potential profile is then derived, is measured along the channel thanks to a CCD camera, which dictates the spatial resolution limit. This technique provides a further important tool to directly visualize the voltage drops at the contacts caused by R_C.

Recently alternative emerging scanning techniques have been proposed, with the aim to map the field[208] and charge density[209] along the channel of a working OFET thanks to confocal microscopy. These electro-optical maps provide useful complementary information about the devices and can be applied also to TG devices, while retaining a high resolution.[210] Their application to the study of contact effects appears promising.

4. Survey of Contact Resistances in Solution Processed OFETs

In this section we report a survey of the contact resistance values as extracted with various methods in OFETs. We restrict the survey to devices where the semiconductor is solution processed. The survey data is collated in **Table 1**, where the semiconductor, OFET architecture, contact material and field-effect mobility are reported together with the extracted R_C values and the extraction method. The record values for contact resistances were mainly obtained in electrolyte-gated transistors, where a ion gel is adopted as a dielectric (**Figure 19**).[211] In TGBC electrolyte-gated OFETs R_C can be as low as 4 Ω·cm in P3HT and 14 Ω·cm in F8T2.[212] These are very impressive values, not attainable when standard dielectric layers are adopted. It is believed that such low R_C, at least three orders of magnitude lower than in non electrolyte-gated OFETs, can be achieved because of the electrochemical doping caused by the migration of ions from the ion-gel into the semiconductor.

Recent literature results on more standard device configurations have been collected in **Figure 20**, where the width normalized R_C values are plotted against μ. Although the literature on the subject is very wide, only a limited set of publications report quantitative and thorough characterizations of R_C. Among these we selected those publications where all the data needed to unambiguously report the width normalized contact resistance and the experimental conditions were clearly presented. Data have been divided in three sets for fairer comparison, on the basis of OFET configuration, and for each class of *p*- and *n*-channel devices. The purpose of Figure 20 is to provide a literature reference rather than extracting a physical property. This is because these data necessarily suffer from various scattering effects, such as different, and not always rigorously applicable, extraction methods, different processing conditions, different regimes for μ calculations, different semiconductor thicknesses, different substrates or different surface treatments and so on. As it is evident from Table 1 for example, the most commonly adopted method is gTLM. This is a perfectly rigorous method as long as it is properly applied, but for example in many reports it is not clear whether R_C is independent of the lateral field

Table 1. Here R_C values in OFETs where the semiconductor was solution processed are collected, together with the indication of the adopted semiconductor, of the device architecture, the contacts material and treatment, if any, the charges mobility and the extraction method. The reference is indicated in the last column. The data were obtained from a survey in the scientific literature in the last twelve years. From the "Method" column, it is evident how TLM is by far the most widely adopted extraction method.

	Semiconductor	Structure	Contacts	μ [$cm^2V^{-1}s^{-1}$]	R_C [Ω·cm]	Method	Ref.
p **channel**							
	TES-ADT	BGTC	Au	0.07	1 x10^6	SKPM	[164]
		BGBC	Au	0.11	5.9 x 10^4		
	P3HT	BGBC	Au/ naphtalene	0.001	1 x 10^6	TLM	[222]
	P3HT	BGBC	Au	0.01	1x 10^4	SKPM	[203]
	P3HT	BGBC	Pt	0.16	1.5 x 10^4	TLM	[223]
	P3HT	BGBC	Au	0.11	1.5 x 10^4	TLM	[156]
		BGBC	Recessed Au	0.15	1.0 x 10^4	TLM	
	P3HT	BGBC	Au	0.15	8.0 x 10^4		[224]
	PTVPhI-Eh	TGBC	Au	0.54	1 x 10^5	TLM	[83]
	F8T2	TGBC	PEDOT:PSS	0.015	1.2 x 10^6	TLM	[225]
	TIPS-p	BGBC	MWCNT:PSS blend	0.13	9 x 10^4	TLM	[148]
	TIPS-p	BGBC	Au	0.20	2.2 x 10^4	SKPM	[205]
	TIPS-p: PS [a]	BGBC	Au	1.01	8.7 x 10^4	SKPM	
	TIPS-p	TGBC	Ag printed	0.06	6 x 10^5	TLM	[172]
			Ag printed/ F4TCNQ	0.9	2 x 10^4		
	TIPS-p	TGBC	Au/ PFDT	0.64	6.5 x 10^5	TLM	[86]
	TIPS-p	TGBC	Au	0.16	1 x 10^6	TLM	[88]
			Au/ PFDT	0.06	2.1 x 10^4		
	pBTTT	TGBC	Au/ PFDT	0.4	4.5 x 10^4	gFPP	[91]
	F8BT [b]	TGBC	Au/ PFDT	0.0018	3 x 10^8	gFPP	[89]
	pBTTT	BGBC	Au	0.56	4.4 x 10^4	TLM	[216]
		BGTC	Au	0.84	6.5 x 10^3		
n **channel**							
	PCBM	BGTC	Ca	0.13	2.2 x 10^3	TLM	[78]
	PCBM	BGBC	Au	0.00035	6 x 10^6	TLM	[226]
	PCBM	BGTC	Al/ TiOx	0.028	4.5 x 10^4	TLM	[227]
	$PC_{71}BM$ [c]	BGTC	Al/ TiOx	0.022	6.0 x 10^4		
	PTVPhI-Eh	TGBC	Au/ CsF	0.26	3.7 x 10^5	TLM	[83]
	N1400	TGBC	Au/ MeTP [d]	0.06	1.4 x 10^5	TLM	[86]
	N1400	TGBC	Au	0.057	3.3 x 10^5	TLM	[88]
			Au/ MeTP	0.07	9.3 x 10^4		
	PDI-8CN2	BGBC	Au/ TP	0.33	1.7 x 10^4	TLM	[90]
	F8BT	TGBC	Au/ 1DT [e]	0.0017	6.7 x 10^8	gFPP	[89]
	P(NDI2OD-T2)	TGBC	Au	0.3	1.1 x 10^4	DM	[75]

[a]TIPS:PS indicates a blend between TIPS-p semiconductor and polystyrene dielectric; [b]F8BT: poly(9,9-di-n-octylfluorene-alt-benzothiadiazole); [c]$PC_{71}BM$: [6,6]-phenyl-C71 butyric acid methyl ester; [d]MeTP: 4-methylthiophenol; [e]1DT: 1-decanethiol.

or not, in which case gTLM is not applicable and leads to an underestimation (see Section 3.1.1).[75] Physical meaning of the proposed plot should be therefore investigated on a controlled set of data, by adopting the same, suitable extraction method, as proposed in a few reports in the literature.[213–215] In Figure 20 an overall scaling can hardly be traced and strong scattering is displayed for high mobility OFETs. It is only by observing the plots for each single dataset that a certain tendency to lower R_C with increasing μ can be observed, due to a better consistency of the data within each set. This is in any case remarkable, suggesting that this behavior, supported by the physics of the injection, is strong enough to emerge from the natural scattering of the data proposed in the literature. Indeed the minimum R_C values for each dataset emerge in the mobility range from 0.1 to

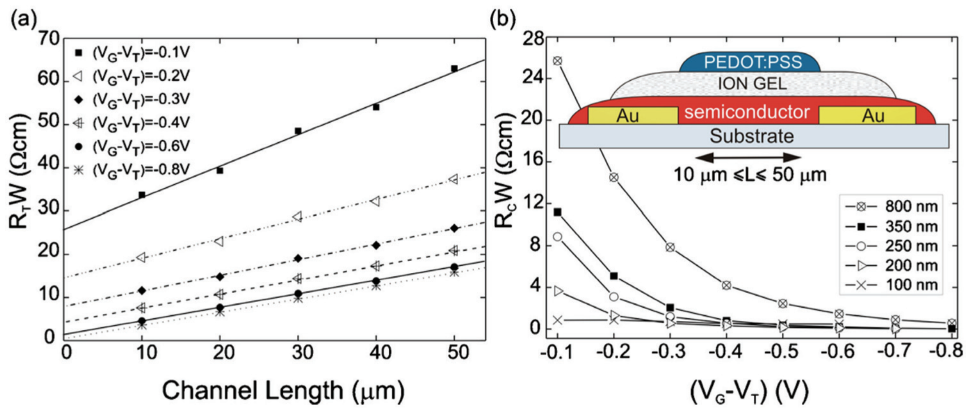

Figure 19. a) Application of the TLM method to extract the width normalized R_C in an electrolyte-gated transistor made with a 800 nm thick P3HT layer, at $V_D = -50$ mV. b) The extracted values are reported as a function of V_G–V_T and for different thicknesses of the P3HT layer. The scheme of the electrolyte-gated P3HT FET is depicted in the inset of panel b. Reproduced with permission.[212] Copyright 2010, American Institute of Physics.

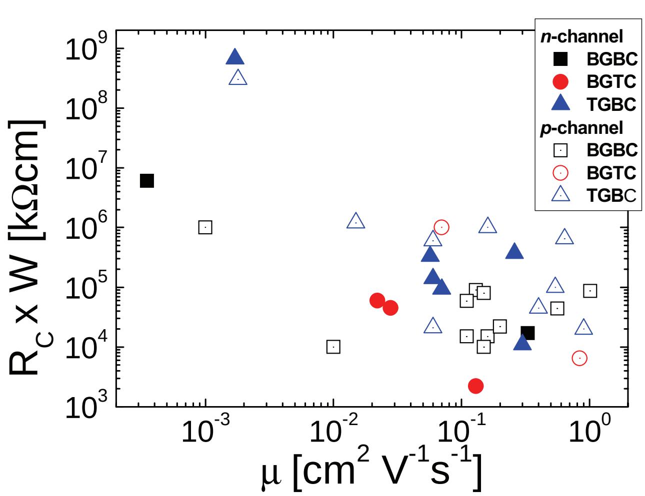

Figure 20. Width normalized R_C values in OFETs, where the semiconductor was solution processed, plotted vs. the device mobility, as obtained from a survey in the scientific literature in the last 12 years. We reported only data from scientific papers where: *i*) R_C x *W* was unambiguously indicated or calculable; *ii*) the extraction method was specified and clearly applied; *iii*) the device mobility was extracted. The data are divided into two main sets (*n*-channel, filled symbols, and *p*-channels, void symbols) and in three subsets based on the device architecture (BGBC, squares, BGTC, circles, TGBC, triangles). The references where the data were sourced from can be found in Table 1.

1 cm^2/Vs, with values in the 1 to 30 kΩ·cm range. The lowest R_C values, below 10 kΩ·cm have so far been obtained with staggered TC architectures. For *p*-type transistors, R_C = 6.5 kΩ·cm was observed for pBTTT devices with Au contacts showing μ = 0.84 cm^2/Vs, where the OTS surface treatment of SiO_2 played the greatest role.[216] In the same report a direct comparison demonstrated the performance improvements achievable with a TC device with respect to the BC case. In the case of *n*-type devices, the lowest R_C of 2.2 kΩ·cm was demonstrated for a PCBM FET with Ca contacts, showing μ = 0.13 cm^2/Vs. In case of the other staggered architecture, TGBC, R_C is still higher than 10 kΩ·cm. TIPS-p devices with Ag/F_4TCNQ contacts show the lowest values in the case of holes, with R_C = 20 kΩ·cm and μ = 0.64 cm^2/Vs, while for electrons the lowest value of 11 kΩ·cm was achieved for a P(NDI2OD-T2) device with Au contacts and μ = 0.30 cm^2/Vs. The latter represents the lowest contact resistance demonstrated so far with bare Au contacts in the BC case. This result is favored by the high bulk conductivity of this polymer, which shows a peculiar face-on orientation, therefore strongly reducing the bulk contribution R_y to the overall resistance.[11,22,75,217–220] Similar record values were observed in BGBC architectures, with R_C = 10 kΩ·cm (μ = 0.15 cm^2/Vs) for *p*-type P3HT FETs with recessed Au contacts and R_C = 17 kΩ·cm (μ = 0.33 cm^2/Vs) in *n*-type N,N'bis-(octyl-)-dicyanoperylene-3,4:9,10-bis(dicarboximide) (PDI-8CN_2) FETs with Au/thiophenol (TP) contacts.

Finally, we provide a rough estimate of the degree of downscaling which is attainable given a certain carrier mobility and contact resistance, and we reconsider survey data under this perspective. We estimate the shortest channel length L_{MIN} below which it is not advantageous to go (as contact resistances start to seriously limit the current), as the channel for which the real current flowing I_{real} is halved with respect to the ideal current I_{ideal} which would flow if contacts were ohmic. To this extent, we can write I_{real} in the linear regime as the product of I_{ideal} times a correction factor due to contact resistances, taking into account that only a fraction of the externally applied V_{DS} drops across the channel: $I_{real} = I_{ideal}\cdot[R_{ch}/(R_{ch} + R_C)]$. L_{MIN} is found by solving $R_{ch} = R_C$. In **Figure 21**, L_{MIN} is plotted versus mobility for various R_C values (straight lines). For correct downscaling below 10 μm with μ in the 0.1–1 cm^2/Vs range, R_C should be smaller than 1 kΩ·cm, whereas for μ in excess of 1 cm^2/Vs, R_C in the 100 Ω·cm range is needed. To give an account of the actual development of the technology, L_{MIN} is also calculated from survey data (note that for the sake of a fair comparison, only values extracted from linear regime are considered here, but R_C in saturation is usually lower and would permit deeper downscaling). Currently, *p*-type OFET downscaling is limited to the deca-micrometer range, with the most notable exception of electrolyte-gated devices which could withstand a sub-micrometer downscaling; whereas for *n*-type OFET L_{MIN} is still larger than 100 μm.

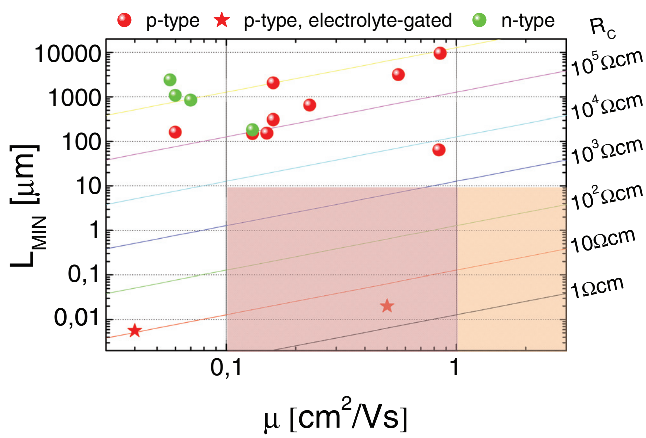

Figure 21. Plot of the channel length L_{MIN} for which the current is halved with respect to the one which would flow in case of ohmic contacts, calculated for various R_C values (given on the right). L_{MIN} is calculated using data from the survey as well: *p*-type standard OFET (red circles), *p*-type electrolyte-gated OFETs (red stars) and *n*-type standard OFET (green circles) Only survey data in the linear regime were considered. For the calculation of R_{ch} a carrier density of 8×10^{12}/ cm^2 was assumed. For L_{MIN} to go below 10 µm with μ in the range 0.1–1 cm^2/Vs, R_C should be no larger than 10 - 1 kΩ cm (pale pink shaded area). Mobilities larger than 1 cm^2/Vs will require R_C at least below 100 Ω·cm (pale orange shaded area).

5. Conclusions

Solution-processed OFETs have the potential for development of a specific industry for low-cost, flexible circuits and for a widespread adoption in many areas, from sensors to microelectronics, if some of the current limitations can be overcome. Here we highlighted how contact issues are in most cases still an open question, which limit performance especially in the case of downscaled devices for higher frequency applications, such as in logic circuits for smart tagging or displays driving.

To study, understand and enhance current injection in OFETs the cooperation of four core fields of knowledge is required: *i*) interface energetics, *ii*) injection physics, *iii*) parameters extraction methods, *iv*) interface engineering.

Interface energetics is still the subject of fundamental studies: this statement is true in general for metal/organic interfaces, and it is even truer in the specific case of OFET interfaces. In addition, metal/semiconductor interfaces in OFETs are inevitably subject to a series of treatments and/or contaminations due to the processing steps needed to develop a device. Detailed studies to ascertain the effects of the OFET process flow on interface energetics are highly desirable.

Regarding current injection, various detailed models which proved to be able to fit specific experimental datasets have been developed, but consensus in the community is still missing. In general, the role and effect of interface disorder is still controversial and only partially accounted for. Specifically in the case of OFETs the effect of the gate on injection has been only partially assessed. Model validity should possibly be checked and demonstrated against various materials, biasing conditions and temperatures. Finally, where injection can be modeled in terms of contact resistances, it is desirable that detailed and complex models are re-cast and/or simplified to yield simple R_C expressions. While keeping in mind that such simplifications will inevitably have a limited application range, this effort is necessary to bridge the work and the world of theorists and models with the one of experimentalists. Indeed, regarding parameter extraction methods especially from current-voltage measurements, it is common practice to rely upon phenomenological approaches and models, from which it is not always straightforward to extract physically relevant parameters. A tighter connection between device modeling and device measuring might foster the development of both fields, and provide the right tools for process engineers in this newly developing industry.

The recent steady improvement of OFETs' field-effect mobility also has a direct effect on the lowering of the contact resistance values in scientific reports, as these two parameters are strongly connected. This does not translate into a direct improvement in charge injection efficiency, as higher mobilities mean lower channel resistances and therefore the relative balance between the charge required by the channel and the charge which the contact can provide does not vary. Only contacts engineering can provide the required breakthrough for the realization of ohmic contacts. For most common architectures, values of ~1 to 10 kΩ·cm represent the very best case, for mobilities in the 0.1 to 1 cm^2/Vs range. In many cases, OFETs reported in the literature can be regarded as contact limited and are not fully suitable for downscaling below a few micrometers. Yet, there are a few examples where short-channel devices were demonstrated with only mild contact effects, and there are also cases, as for electrolyte-gated transistors, where extremely low resistances were achieved, indicating that improvements in this direction are achievable.

It is foreseeable that many more chemical and surface treatments for contacts will be demonstrated in the short term, methods that will have to be tested against device stability and against compatibility with low thermal budget and solution-based processes. These achievements will concur to the development of high-performance, fast-switching OFETs fabrication processes compatible with roll-to-roll, large area electronic circuit manufacturing.

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